



❑ યાદ રાખો :

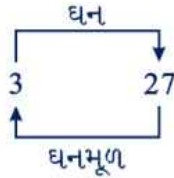
⇒ 1 થી 10 ના ઘન

|                       |                         |
|-----------------------|-------------------------|
| $1^3 \rightarrow 1$   | $6^3 \rightarrow 216$   |
| $2^3 \rightarrow 8$   | $7^3 \rightarrow 343$   |
| $3^3 \rightarrow 27$  | $8^3 \rightarrow 512$   |
| $4^3 \rightarrow 64$  | $9^3 \rightarrow 729$   |
| $5^3 \rightarrow 125$ | $10^3 \rightarrow 1000$ |

(4)  $\sqrt{27 \div \sqrt{27 \div \sqrt{27 \div \dots \infty}}}$   
 (A) 2 (B) 3 (C)  $\sqrt{3}$  (D)  $\sqrt[3]{3}$

**Solution :**

⇒ 27 નું ઘનમૂળ = 3



∴ (B) 3

❑ સરવાળા - બાદબાકી :

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots \infty}}}} = ?$$

⇒ અહીં, જે સંખ્યા આપેલી હોય તેનો કમિક સંખ્યાનો ગુણાકાર શોધવો.

$$\therefore 2 \rightarrow 1 \times 2$$

⇒ આપેલ સંખ્યાના બે કમિક અવયવો કે જેના ગુણાકાર સંખ્યા અને તફાવત 1 થાય.

⇒ રકમમાં સરવાળાની નિશાની (+) આપેલ હોય તો મોટી સંખ્યા જવાબ બનશે.

$$\therefore 2 \rightarrow 1 \times 2 \rightarrow \text{મોટી સંખ્યા જવાબ ગણાશે.}$$

$$= 2$$

❖  $\sqrt{2 - \sqrt{2 - \sqrt{2 - \sqrt{2 - \dots \infty}}}} = ?$

⇒ જો  $\infty$  આપેલ હોય ત્યારે ઉપયુક્ત ક્રમાનુસાર જ આપેલ સંખ્યાનો કમિક ગુણાકાર શોધવો.

$$\therefore 2 \rightarrow 1 \times 2$$

⇒ જો રકમમાં બાદબાકીની (-) નિશાની આપેલ હોય તો નાની સંખ્યા જવાબ આવશે.

$$\Rightarrow \text{જેમ કે, } 2 \rightarrow 1 \times 2$$



નાની સંખ્યા જવાબ મળે = 1

(5)  $\sqrt{12 + \sqrt{12 + \sqrt{12 + \dots \infty}}}$

(A) 4 (B) 3 (C) 2 (D) 12

**Solution :**

$$\Rightarrow \sqrt{12 + \sqrt{12 + \sqrt{12 + \dots \infty}}}$$

3 × 4

⇒ સરવાળાની નિશાની હોવાથી મોટી સંખ્યા જવાબ મળે.

∴ (A) 4

(6)  $\sqrt{30 - \sqrt{30 - \sqrt{30 - \dots \infty}}}$

(A) 5 (B) 6 (C) 7 (D) 30

**Solution :**

$$\sqrt{30 - \sqrt{30 - \sqrt{30 - \dots \infty}}}$$

5 × 6

⇒ બાદબાકીની નિશાની હોવાથી જવાબ નાની સંખ્યા આવશે. તેથી જવાબ 5 આવે.

∴ (A) 5

## Type # 2

❑ સમજૂતી :

⇒ ચાર પગવાળા પ્રાણી =  $\frac{\text{પગ}}{2}$  - માથા

⇒ બે પગવાળા પ્રાણી = કુલ માથા - ચાર પગવાળા પ્રાણી

(7) વિરાટ પાસે કેટલાક સસલા અને પોપટ છે. જેમાં માથા 45 છે જ્યારે પગ 130 છે તો વિરાટ પાસે કુલ કેટલા પોપટ હશે ?

(A) 35 (B) 15 (C) 25 (D) 30

**Solution :**

⇒ સસલા અને પોપટને માથુ એક હોય એટલે બંનેની સંખ્યા કુલ 45 છે

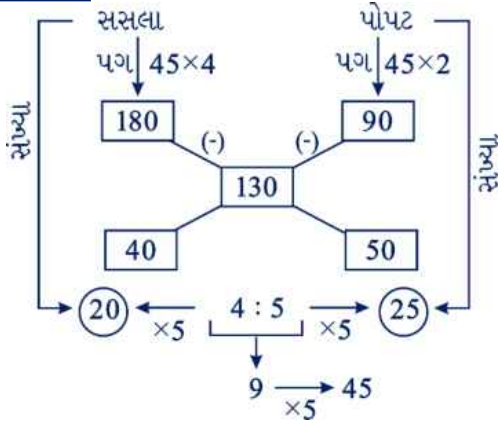
⇒ રીત : 1

$$\begin{aligned} \therefore \text{ચાર પગ} &= \frac{\text{પગ}}{2} - \text{માથા} \\ &= \frac{130}{2} - 45 \\ &= 65 - 45 \end{aligned}$$

∴ ચાર પગ = 20 → સસલાની સંખ્યા મળી.

∴ બે પગ = 45 - 20 = 25 → પોપટની સંખ્યા.

⇒ રીત : 2



- (8) રોકી પાસે કેટલાક મોર અને હાથી છે. જો મોર અને હાથીનાં કુલ માથા 20 હોય અને કુલ પગ 56 હોય તો હાથીની સંખ્યા કેટલી ?

(A) 12 (B) 6 (C) 7 (D) 8

Solution :

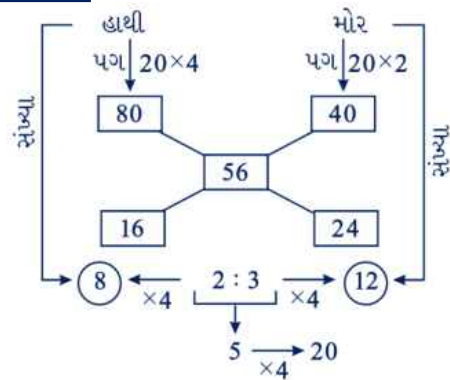
⇒ રીત : 1

$$\begin{aligned}\therefore \text{ચાર પગ} &= \frac{\text{પગ}}{2} - \text{માથા} \\ &= \frac{56}{2} - 20 \\ &= 28 - 20\end{aligned}$$

$\therefore$  ચાર પગ = 8  $\rightarrow$  હાથીની સંખ્યા

$\therefore$  બે પગ = 20 - 8 = 12  $\rightarrow$  મોરની સંખ્યા

⇒ રીત : 2



- (9) પુષ્પા પાસે કેટલાક બતક અને ગાય છે. જો બતક અને ગાય ના કુલ માથા 30 અને કુલ પગ 90 હોય તો બતકની સંખ્યા કેટલી ?

(A) 15 (B) 20 (C) 25 (D) 30

Solution :

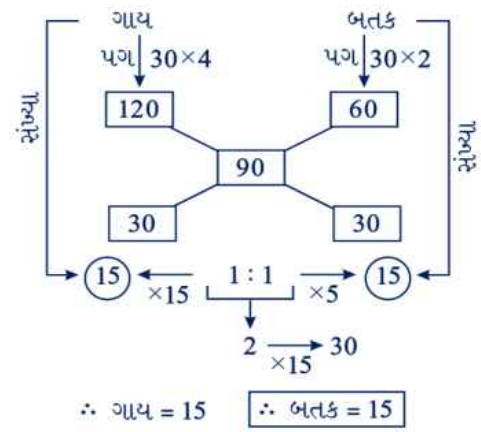
⇒ રીત : 1

$$\begin{aligned}\therefore \text{ચાર પગ} &= \frac{\text{પગ}}{2} - \text{માથા} \\ &= \frac{90}{2} - 30 \\ &= 45 - 30\end{aligned}$$

$\therefore$  ચાર પગ = 15  $\rightarrow$  ગાયની સંખ્યા

$\therefore$  બે પગ = 30 - 15 = 15  $\rightarrow$  બતકની સંખ્યા

⇒ રીત : 2



- (10) ડોનાલ્ડ ટ્રમ્પ પાસે કેટલીક મરઘી અને ભેંસ છે. મરઘી અને ભેંસના કુલ માથા 18 હોય અને પગ 48 હોય તો મરઘીની સંખ્યા કેટલી ?

(A) 10 (B) 16 (C) 18 (D) 12

Solution :

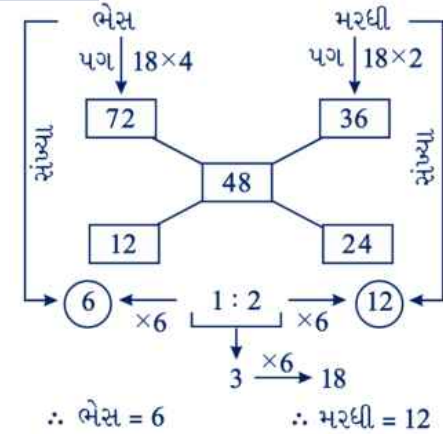
⇒ રીત : 1

$$\begin{aligned}\therefore \text{ચાર પગ} &= \frac{\text{પગ}}{2} - \text{માથા} \\ &= \frac{48}{2} - 18 \\ &= 24 - 18\end{aligned}$$

$\therefore$  ચાર પગ = 6  $\rightarrow$  ભેંસની સંખ્યા

$\therefore$  બે પગ = 18 - 6 = 12  $\rightarrow$  મરઘીની સંખ્યા

⇒ રીત : 2



- (11) અનુષ્ઠા પાસે કેટલાક મોર અને કેટલાક બળદ છે જો મોર અને બળદના કુલ માથા 40 હોય અને કુલ પગ 114 હોય તો બળદની કુલ સંખ્યા કેટલી ?

(A) 8 (B) 14 (C) 17 (D) 20

**Solution :**

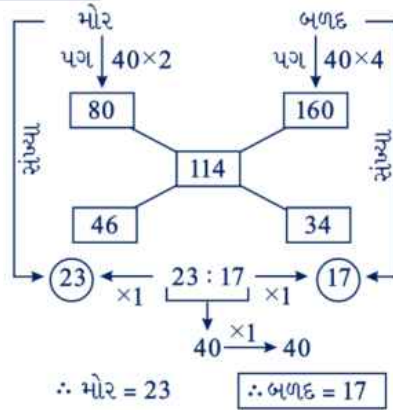
⇒ ચિત્ર : 1

$$\begin{aligned}\therefore \text{ચાર પગ} &= \frac{\text{પગ}}{2} - \text{માથા} \\ &= \frac{114}{2} - 40 \\ &= 57 - 40\end{aligned}$$

$$\therefore \text{ચાર પગ} = 17 = \text{બળદની સંખ્યા}$$

$$\therefore \text{બે પગ} = 40 - 17 = 23 \text{ મોરની સંખ્યા}$$

⇒ ચિત્ર : 2



$$\therefore \text{મોર} = 23$$

$$\therefore \text{બળદ} = 17$$

∴ (C) 17

### Type # 3

□ સમજૂતી :

$$\Rightarrow \text{કુલ મેચ (ઘટના)} = \frac{n(n-1)}{2}$$

- (12) બેડમિન્ટનની એક રમતમાં 5 ખેલાડી ભાગ લઈ રહ્યા છે. દરેક ખેલાડી અન્ય દરેક ખેલાડી સાથે એક મેચ રમે તો કુલ કેટલી મેચ રમવી પડે ?

(A) 12 (B) 16 (C) 18 (D) 10

**Solution :**

$$\Rightarrow n = 5$$

$$\begin{aligned}\Rightarrow \text{કુલ મેચ (ઘટના)} &= \frac{n(n-1)}{2} \\ &= \frac{5(5-1)}{2} \\ &= \frac{5 \times 4}{2} \\ &= \frac{20}{2} \\ &= 10\end{aligned}$$

∴ (D) 10

- (13) એક લગ્ન સમારંભમાં 10 વ્યક્તિઓ ભેગા થાય છે. દરેક વ્યક્તિ અન્ય દરેક વ્યક્તિ સાથે હાથ મિલાવે તો હાથ મિલાવવાની કુલ સંખ્યા કેટલી ?

(A) 30 (B) 45

(C) 25 (D) 40

**Solution :**

$$\Rightarrow n = 10$$

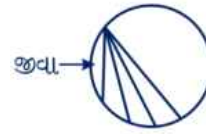
$$\begin{aligned}\Rightarrow \text{હાથ મિલાવવાની કુલ સંખ્યા} &= \frac{n(n-1)}{2} \\ &= \frac{10(10-1)}{2} \\ &= \frac{5}{2} \times 9 \\ &= \frac{10 \times 9}{2} \\ &= 45\end{aligned}$$

∴ (B) 45

- (14) વર્તુળના પરિઘ પર 12 બિંદુઓ આવેલા છે. દરેક બિંદુમાંથી અન્ય બિંદુમાંથી અન્ય બિંદુ તરફ જવા દોરવામાં આવે તો જવાની કુલ સંખ્યા શોધો ?

(A) 66 (B) 36 (C) 46 (D) 52

**Solution :**



$$\begin{aligned}\Rightarrow \text{જવાની સંખ્યા} &= \frac{n(n-1)}{2} \\ &= \frac{12(12-1)}{2} \\ &= 6 \times 11 \\ &= 66\end{aligned}$$

∴ (A) 66

- (15) બેડમિન્ટનની એક રમતમાં કુલ 28 મેચ રમવી પડે છે. તો દરેક ખેલાડી અન્ય ખેલાડી સાથે મેચ રમે તો કુલ કેટલા ખેલાડીઓએ રમતમાં ભાગ લીધો હશે ?

(A) 5 (B) 7 (C) 8 (D) 6

**Solution :**

$$\begin{aligned}\Rightarrow \text{કુલ મેચ (ઘટના)} &= \frac{n(n-1)}{2} \\ 28 &= \frac{n(n-1)}{2} \\ 28 \times 2 &= n(n-1) \\ 56 &= n(n-1)\end{aligned}$$



⇒ (∵  $n(n-1)$  એ કમિક સંખ્યા છે.)

એટલે, 56 ની કમિક સંખ્યા =  $8 \times 7 = 56$

નો ગુણાકાર ↓ ↓

$$= n(n-1) \text{ થાય}$$

$$\therefore 56 = 8 \times 7$$

$$\therefore n = 8$$

∴ (C) 8

(16) બેડમિન્ટનની એક રમતમાં કુલ 21 મેચ રમવી પડે છે. દરેક અન્ય ખેલાડી સાથે એક મેચ રમે તો કુલ કેટલા ખેલાડીઓએ રમતમાં ભાગ લીધો હશે ?

(A) 6 (B) 7 (C) 9 (D) 8

**Solution :**

$$\Rightarrow \text{કુલ મેચ (ઘટન)} = \frac{n(n-1)}{2}$$

$$21 = \frac{n(n-1)}{2}$$

$$21 \times 2 = n(n-1)$$

$$42 = 7 \times 6 \text{ (કમિક સંખ્યાનો ગુણાકાર)}$$

[∵  $n(n-1)$  એ કમિક સંખ્યા છે.]

$$n = 7$$

∴ (B) 7

### Type # 4

□ સમજૂતી :

⇒ ભેટની (Gift) સંખ્યા =  $n(n-1)$

(17) 9 વ્યક્તિઓ એક મિટિંગમાં મળે છે. દરેક વ્યક્તિ એકબીજાને ભેટ આપે છે તો ભેટની સંખ્યા શોધો.

(A) 72 (B) 36 (C) 48 (D) 38

**Solution :**

$$\Rightarrow \text{ભેટની સંખ્યા} = n(n-1) = 9 \times (9-1)$$

$$= 9 \times 8$$

$$= 72$$

∴ (A) 72

(18) 11 વ્યક્તિઓ એક મિટિંગમાં મળે છે. દરેક વ્યક્તિ એકબીજાને ભેટ આપે છે તો ભેટ સંખ્યા શોધો.

(A) 110 (B) 90 (C) 54 (D) 55

**Solution :**

$$\Rightarrow \text{ભેટની સંખ્યા} = n(n-1) = 11 \times (11-1)$$

$$= 11 \times 10$$

$$= 110$$

∴ (A) 110

(19) 4 વ્યક્તિઓ એક મિટિંગમાં મળે છે. દરેક વ્યક્તિ એકબીજાને ભેટ આપે છે તો ભેટની સંખ્યા શોધો.

(A) 16 (B) 12 (C) 6 (D) 24

**Solution :**

$$\Rightarrow \text{ભેટની સંખ્યા} = n(n-1) = 4 \times (4-1)$$

$$= 4 \times 3$$

$$= 12$$

∴ (B) 12

(20) 6 વ્યક્તિઓ એક મિટિંગમાં મળે છે. દરેક વ્યક્તિ એકબીજાને ભેટ આપે છે તો ભેટની સંખ્યા શોધો.

(A) 10 (B) 40 (C) 30 (D) 20

**Solution :**

$$\Rightarrow \text{ભેટની સંખ્યા} = n(n-1) = 6 \times (6-1)$$

$$= 6 \times 5$$

$$= 30$$

∴ (C) 30

(21) એક મિટિંગમાં અમુક વ્યક્તિઓ મળે છે. દરેક વ્યક્તિ એકબીજાને ભેટ આપે છે જો ભેટની કુલ સંખ્યા 56 હોય તો વ્યક્તિઓની સંખ્યા શોધો.

(A) 6 (B) 7 (C) 9 (D) 8

**Solution :**

$$\Rightarrow \text{કુલ ભેટ} = n(n-1)$$

$$56 = n(n-1) \text{ (અહીં બંને કમિક સંખ્યા છે.)}$$

$$56 = 8 \times 7 \text{ (કમિક સંખ્યાનો ગુણાકાર છે.)}$$

$$n = 8$$

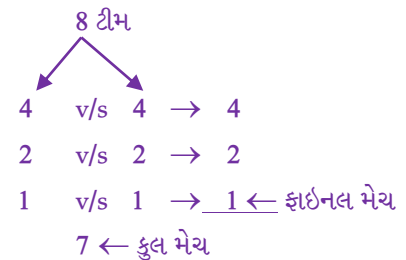
∴ (D) 8

### Type # 5

□ સમજૂતી :

⇒ જ્યારે મેચ નોક આઉટ રમાવાની હોય (એટલે કે હારે તે ઘરે) ત્યારે . . . . .

⇒ Ex.



$$\Rightarrow \text{કુલ મેચ} = n-1$$

$$= 8-1$$

$$= 7$$

- (22) 32 ટીમને બે ભાગમાં વહેંચી નોક આઉટ પદ્ધતિથી મેચ રમાડવામાં આવે તો ફાઈનલ વિજેતા નક્કી માટે કુલ કેટલી મેચ રમાડવી પડે ?

(A) 31 (B) 24  
(C) 37 (D) 27

**Solution :**

$$\begin{aligned}\Rightarrow \text{કુલ મેચ} &= n - 1 \\ &= 32 - 1 \\ &= 31\end{aligned}$$

$\therefore$  (A) 31

- (23) 16 ટીમને બે ભાગમાં વહેંચી નોક આઉટ પદ્ધતિથી મેચ રમાડવામાં આવે તો ફાઈનલ વિજેતા નક્કી માટે કુલ કેટલી મેચ રમાડવી પડે ?

(A) 17 (B) 21  
(C) 15 (D) 24

**Solution :**

$$\begin{aligned}\Rightarrow \text{કુલ મેચ} &= n - 1 \\ &= 16 - 1 \\ &= 15\end{aligned}$$

$\therefore$  (C) 15

- (24) 64 ટીમને બે ભાગમાં વહેંચી નોક આઉટ પદ્ધતિથી મેચ રમાડવામાં આવે તો ફાઈનલ વિજેતા નક્કી માટે કુલ કેટલી મેચ રમાડવી પડે ?

(A) 53 (B) 84  
(C) 73 (D) 63

**Solution :**

$$\begin{aligned}\Rightarrow \text{કુલ મેચ} &= n - 1 \\ &= 64 - 1 \\ &= 63\end{aligned}$$

$\therefore$  (D) 63

- (25) 22 ટીમને બે ભાગમાં વહેંચી નોક આઉટ પદ્ધતિથી મેચ રમાડવામાં આવે તો ફાઈનલ વિજેતા નક્કી માટે કુલ કેટલી મેચ રમાડવી પડે ?

(A) 19 (B) 21 (C) 17 (D) 23

**Solution :**

$$\begin{aligned}\Rightarrow \text{કુલ મેચ} &= n - 1 \\ &= 22 - 1 \\ &= 21\end{aligned}$$

$\therefore$  (B) 21

- (26) અમુક ટીમને બે ભાગમાં વહેંચી નોક આઉટ પદ્ધતિથી મેચ રમાડવામાં આવે તો ફાઈનલ વિજેતા નક્કી માટે કુલ 7 મેચ રમાડવી પડે છે તો ટીમની સંખ્યા શોધો.

(A) 6 (B) 7 (C) 9 (D) 8

**Solution :**

$$\begin{aligned}\Rightarrow \text{કુલ મેચ} &= n - 1 \\ \therefore 7 &= n - 1 \\ \therefore n &= 7 + 1 \\ \therefore n &= 8\end{aligned}$$

$\therefore$  (D) 8

### Type # 6

□ સમજૂતી :

$$\therefore 10 \div 10 \div 10 \div 10 \div 10$$

$$= 10 \times \frac{1}{10} \times \frac{1}{10} \times \frac{1}{10} \times \frac{1}{10}$$

[ $\therefore$  ભાગાકારનો ગુણાકાર કરવો પછીની સંખ્યાનો વ્યસ્ત લેવો.]

$$= \frac{1}{10 \times 10 \times 10}$$

$$= \frac{1}{1000}$$

$$= 0.001$$

$$= \frac{1}{10^3}$$

$$= 10^{-3}$$

□ યાદ રાખો :

$\Rightarrow$  1 થી 16 ના ઘન

$$1^3 \rightarrow 1 \quad 6^3 \rightarrow 216 \quad 11^3 \rightarrow 1331$$

$$2^3 \rightarrow 8 \quad 7^3 \rightarrow 343 \quad 12^3 \rightarrow 1728$$

$$3^3 \rightarrow 27 \quad 8^3 \rightarrow 512 \quad 13^3 \rightarrow 2197$$

$$4^3 \rightarrow 64 \quad 9^3 \rightarrow 729 \quad 14^3 \rightarrow 2744$$

$$5^3 \rightarrow 125 \quad 10^3 \rightarrow 1000 \quad 15^3 \rightarrow 3375$$

$$16^3 \rightarrow 4096$$

- (27) સાદુંરૂપ આપો :  $\frac{1}{12} \div \frac{1}{12} \div \frac{1}{12} \div \frac{1}{12} \div \frac{1}{12}$

(A) 1287

(B) 1278

(C) 1728

(D) 1782

**Solution :**

$$\Rightarrow \frac{1}{12} \times 12 \times 12 \times 12 \times 12 = 12^3 = 1728$$

$\therefore$  (C) 1728

- (28) સાદુરૂપ આપો :  $\frac{1}{13} \div \frac{1}{13} \div \frac{1}{13} \div \frac{1}{13} \div \frac{1}{13}$   
 (A) 2197 (B) 1444 (C) 1331 (D) 2917

**Solution :**

$$\Rightarrow \frac{1}{13} \times \cancel{13} \times 13 \times 13 \times 13 = 13^3 = 2197$$

$\therefore$  (A) 2197

- (29) સાદુરૂપ આપો :  $\frac{1}{10} \div \frac{1}{10} \div \frac{1}{10} \div \frac{1}{10} \div \frac{1}{10}$   
 (A) 800 (B) 1000 (C) 1200 (D) 900

**Solution :**

$$\Rightarrow \frac{1}{10} \times \cancel{10} \times 10 \times 10 \times 10 = 10^3 = 1000$$

$\therefore$  (B) 1000

- (30) સાદુરૂપ આપો :  $\frac{1}{8} \div \frac{1}{8} \div \frac{1}{8} \div \frac{1}{8} \div \frac{1}{8}$   
 (A) 1 (B) 8 (C) 64 (D) 512

**Solution :**

$$\Rightarrow \frac{1}{8} \times \cancel{8} \times 8 \times 8 \times 8 = 8^3 = 512$$

$\therefore$  (D) 512

- (31) સાદુરૂપ આપો :  $\frac{1}{5} \div \frac{1}{5} \div \frac{1}{5} \div \frac{1}{5} \div \frac{1}{5}$   
 (A) 5 (B) 25 (C) 125 (D) 1

**Solution :**

$$\Rightarrow \frac{1}{5} \times \cancel{5} \times 5 \times 5 \times 5 = 5^3 = 125$$

$\therefore$  (C) 125

### Type # 7

□ સમજૂતી :

$$\Rightarrow (a+b)^2 = (a+b)(a+b) = a^2 + 2ab + b^2$$

$$\Rightarrow (a-b)^2 = (a-b)(a-b) = a^2 - 2ab + b^2$$

$$\Rightarrow a^2 - b^2 = (a+b)(a-b)$$

$$\Rightarrow a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$= (a+b)^3 - 3ab(a+b)$$

$$\Rightarrow a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$= (a-b)^3 + 3ab(a-b)$$

$$\Rightarrow \frac{a^3 + b^3}{a^2 - ab + b^2} = a+b$$

$$\Rightarrow \frac{a^3 - b^3}{a^2 + ab + b^2} = \frac{(a-b)(\cancel{a^2 + ab + b^2})}{(\cancel{a^2 + ab + b^2})} = a-b$$

$$\Rightarrow \frac{a^2 - b^2}{a+b} = \frac{(\cancel{a+b})(a-b)}{(\cancel{a+b})} = a-b$$

$$\Rightarrow \frac{a^2 - b^2}{a-b} = \frac{(a+b)(\cancel{a-b})}{(\cancel{a-b})} = a+b$$

$$\Rightarrow (a+b)^2 + (a-b)^2$$

$$= a^2 + \cancel{2ab} + b^2 + a^2 - \cancel{2ab} + b^2$$

$$= 2a^2 + 2b^2$$

$$= 2(a^2 + b^2)$$

$$\Rightarrow \frac{(a+b)^2 + (a-b)^2}{a^2 + b^2} = \frac{2(\cancel{a^2 + b^2})}{\cancel{a^2 + b^2}} = 2$$

$$\Rightarrow (a+b)^2 - (a-b)^2$$

$$= \cancel{a^2} + 2ab + \cancel{b^2} - \cancel{a^2} + 2ab - \cancel{b^2}$$

$$= 4ab$$

$$\Rightarrow \frac{(a+b)^2 - (a-b)^2}{ab} = \frac{4ab}{ab} = 4$$

- (32) સાદુરૂપ આપો :  $\frac{5.2 \times 5.2 - 2.7 \times 2.7}{7.9 \times 2.5}$

(A) 1.3 (B) 1 (C) 1.9 (D) 0.3

**Solution :**

$$\Rightarrow a = 5.2, b = 2.7$$

$$\Rightarrow a+b = 5.2 + 2.7 = 7.9$$

$$a-b = 5.2 - 2.7 = 2.5$$

$$\Rightarrow a \times a = a^2$$

$$a + a = 2a$$

$$\therefore \frac{a^2 - b^2}{(a+b)(a-b)} = \frac{(a+b)(\cancel{a-b})}{(\cancel{a+b})(a-b)} = 1$$

$\therefore$  (B) 1

- (33) સાદુરૂપ આપો :  $\frac{5.2 \times 5.2 - 2.7 \times 2.7}{7.9}$

(A) 2.5 (B) 1 (C) 1.9 (D) 0.3

**Solution :**

$$\Rightarrow a = 5.2, b = 2.7$$

$$\Rightarrow a+b = 5.2 + 2.7 = 7.9$$

$$\therefore \frac{a^2 - b^2}{a+b} = \frac{(\cancel{a+b})(a-b)}{(\cancel{a+b})} = a-b$$

$$\therefore a-b = 5.2 - 2.7$$

$$= 2.5$$

$\therefore$  (A) 2.5

- (34)  $\frac{7.35 \times 7.35 - 2.25 \times 2.25}{0.24}$   
 (A) 204 (B) 320 (C) 225 (D) 304

**Solution :**

$$\begin{aligned} \Rightarrow a &= 7.35 & b &= 2.25 \\ \therefore \frac{a^2 - b^2}{0.24} &= \frac{a^2 - b^2}{0.24} \\ \therefore \frac{(a+b)(a-b)}{0.24} &= \frac{(7.35+2.25)(7.35-2.25)}{0.24} \\ &= \frac{9.6 \times 5.1}{0.24} \\ &= \frac{49.68}{0.24} \\ &= 207 \\ \therefore (A) 204 \end{aligned}$$

- (35)  $\frac{(379+276)^2 + (379-276)^2}{379 \times 379 + 276 \times 276}$   
 (A) 2 (B) 655 (C) 103 (D) 1

**Solution :**

$$\begin{aligned} \Rightarrow a &= 379 & b &= 276 \\ \therefore \frac{(a+b)^2 + (a-b)^2}{a^2 + b^2} &= \frac{(379+276)^2 + (379-276)^2}{379 \times 379 + 276 \times 276} \\ &= \frac{a^2 + b^2 + a^2 + b^2}{a^2 + b^2} \\ &= \frac{2(a^2 + b^2)}{a^2 + b^2} \\ &= 2 \\ \therefore (A) 2 \end{aligned}$$

- (36) સાદુંરૂપ આપો :  $\frac{(326+222)^2 - (326-222)^2}{(326 \times 222)}$   
 (A) 1 (B) 4 (C) 3 (D) 2

**Solution :**

$$\begin{aligned} \Rightarrow a &= 326 & b &= 222 \\ \therefore \frac{(a+b)^2 - (a-b)^2}{a \times b} &= \frac{(326+222)^2 - (326-222)^2}{(326 \times 222)} \\ &= \frac{4ab}{ab} \\ &= 4 \\ \therefore (B) 4 \end{aligned}$$

- (37)  $\frac{(589+187)^2 - (589-187)^2}{589 \times 187}$   
 (A) 4 (B) 766 (C) 402 (D)  $\frac{201}{388}$

**Solution :**

$$\begin{aligned} \Rightarrow a &= 589 & b &= 187 \\ &= \frac{(589+187)^2 - (589-187)^2}{(589 \times 187)} \\ &= \frac{(a+b)^2 - (a-b)^2}{a \times b} \\ &= \frac{4ab}{ab} \\ &= 4 \\ \therefore (A) 4 \end{aligned}$$

- (38) સાદુંરૂપ આપો :  $\frac{2.31 \times 2.31 + 2 \times 2.31 \times 1.69 + 1.69 \times 1.69}{2.31 + 1.69}$   
 (A) 3 (B) 4 (C) 2 (D) 5

**Solution :**

$$\begin{aligned} \Rightarrow a &= 2.31 & b &= 1.69 \\ \therefore \frac{a^2 + 2ab + b^2}{a+b} &= \frac{2.31 \times 2.31 + 2 \times 2.31 \times 1.69 + 1.69 \times 1.69}{2.31 + 1.69} \\ &= \frac{(a+b)^2}{a+b} \\ &= \frac{(a+b) \times (a+b)}{(a+b)} \\ &= a+b \\ &= 2.31 + 1.69 \\ &= 4 \\ \therefore (B) 4 \end{aligned}$$

- (39) સાદુંરૂપ આપો :  $\frac{0.2 \times 0.2 \times 0.2 + 0.5 \times 0.5 \times 0.5}{0.02 \times 0.02 \times 0.02 + 0.05 \times 0.05 \times 0.05}$   
 (A) 5000 (B) 10000  
 (C) 100 (D) 1000

**Solution :**

$$\begin{aligned} \therefore \frac{\overbrace{0.2 \times 0.2 \times 0.2}^{3 \text{ દશાંશ}} + \overbrace{0.5 \times 0.5 \times 0.5}^{3 \text{ દશાંશ}}}{\underbrace{0.02 \times 0.02 \times 0.02}_{6 \text{ દશાંશ}} + \underbrace{0.05 \times 0.05 \times 0.05}_{6 \text{ દશાંશ}}} \\ = \frac{\cancel{0.2} + \cancel{0.5} \text{ દશાંશ}}{\cancel{0.02} + \cancel{0.05} \text{ દશાંશ}} \\ = \frac{2}{5} \end{aligned}$$



$$= \frac{1}{1000}$$

[જેટલા દશાંશ અંતે વધે તે અંશમાં જતા જવાબ બનશે.]

$$\therefore (D) 1000$$

$$(40) \frac{(0.0002)^2 + (0.0033)^2 + (0.0444)^2}{(0.00002)^2 + (0.00033)^2 + (0.00444)^2} = ?$$

(A) 100 (B) 80 (C) 90 (D) 110

**Solution :**

$$\therefore \frac{\frac{4}{(0.0002)^2} + \frac{4}{(0.0033)^2} + \frac{4}{(0.0444)^2}}{\frac{4}{(0.00002)^2} + \frac{4}{(0.00033)^2} + \frac{4}{(0.00444)^2}}$$

[જ્યારે દશાંશનો વર્ગ કરવામાં આવે ત્યારે દશાંશ બે ગણા થાય.]

$$\therefore \frac{8}{10} \text{ 2 વધ્યા એટલે } \frac{1}{100} = 100$$

$$\therefore (A) 100$$

### Type # 8

□ યાદ રાખો :

⇒ 1 થી 30 ના વર્ગ

|                        |                        |                        |
|------------------------|------------------------|------------------------|
| $1^2 \rightarrow 1$    | $11^2 \rightarrow 121$ | $21^2 \rightarrow 441$ |
| $2^2 \rightarrow 4$    | $12^2 \rightarrow 144$ | $22^2 \rightarrow 484$ |
| $3^2 \rightarrow 9$    | $13^2 \rightarrow 169$ | $23^2 \rightarrow 529$ |
| $4^2 \rightarrow 16$   | $14^2 \rightarrow 196$ | $24^2 \rightarrow 576$ |
| $5^2 \rightarrow 25$   | $15^2 \rightarrow 225$ | $25^2 \rightarrow 625$ |
| $6^2 \rightarrow 36$   | $16^2 \rightarrow 256$ | $26^2 \rightarrow 676$ |
| $7^2 \rightarrow 49$   | $17^2 \rightarrow 289$ | $27^2 \rightarrow 729$ |
| $8^2 \rightarrow 64$   | $18^2 \rightarrow 324$ | $28^2 \rightarrow 784$ |
| $9^2 \rightarrow 81$   | $19^2 \rightarrow 361$ | $29^2 \rightarrow 841$ |
| $10^2 \rightarrow 100$ | $20^2 \rightarrow 400$ | $30^2 \rightarrow 900$ |

(41) x ની કિંમત શોધો.

$$\sqrt{(189-x)} = \sqrt{(178-\sqrt{81})}$$

(A) 20 (B) 27 (C) 38 (D) 49

**Solution :**

$$\Rightarrow \sqrt{(189-x)} = \sqrt{(178-\sqrt{81})}$$

[બંને બાજુ વર્ગ કરતા વર્ગમૂળની નિશાની નીકળી જાય.]

$$189 - x = 178 - \sqrt{81}$$

$$189 - x = 178 - 9$$

$$189 - x = 169$$

$$x = 189 - 169$$

$$x = 20$$

$$\therefore (A) 20$$

(42) x ની કિંમત શોધો :

$$\sqrt{(157-x)} = \sqrt{(155-\sqrt{121})}$$

(A) 13 (B) 16 (C) 17 (D) 14

**Solution :**

$$\therefore 157 - x = 155 - \sqrt{121}$$

$$157 - x = 155 - 11$$

$$157 - x = 144$$

$$x = 157 - 144$$

$$x = 13$$

$$\therefore (A) 13$$

(43) x ની કિંમત શોધો :

$$\sqrt{(430-x)} = \sqrt{(86-\sqrt{25})}$$

(A) 729 (B) 372 (C) 298 (D) 349

**Solution :**

$$\Rightarrow 430 - x = 86 - \sqrt{25}$$

$$430 - x = 86 - 5$$

$$430 - x = 81$$

$$x = 430 - 81$$

$$x = 349$$

$$\therefore (D) 349$$

(44) x ની કિંમત શોધો :

$$\sqrt{441} \div 21 + \sqrt{484} = 1 \times x$$

(A) 26 (B) 25 (C) 23 (D) 24

**Solution :**

$$\Rightarrow \sqrt{441} = 21$$

$$\Rightarrow \sqrt{484} = 22$$

$$\Rightarrow \frac{21}{21} + 22 = x$$

$$1 + 22 = x$$

$$x = 23$$

$$\therefore (C) 23$$





(55)  $x$  ની કિંમત શોધો :

$$\sqrt{121} \div 11 + \sqrt{144} = 1 \times x$$

(A) 16 (B) 14 (C) 13 (D) 15

Solution :

$$\therefore \frac{11}{11} + 12 = x$$

$$\therefore 1 + 12 = x$$

$$\therefore x = 13$$

$$\therefore (C) 13$$

(56)  $x$  ની કિંમત શોધો :

$$\sqrt{361} \div 19 + \sqrt{400} = 1 \times x$$

(A) 24 (B) 22 (C) 23 (D) 21

Solution :

$$\therefore \frac{19}{19} + 20 = x$$

$$\therefore 1 + 20 = x$$

$$\therefore x = 21$$

$$\therefore (D) 21$$

## Type # 9

□ યાદ રાખો :

⇒ જો એકમના અંક તરીકે 0, 1, 4, 5, 6, 9 હોય તો તેનો ઘન કરો તો તેની તે જ સંખ્યા આવે.

⇒ ઘનમૂળ કાઢો તો પણ તે જ સંખ્યા મળે.

|    |   |      |       |
|----|---|------|-------|
| 1  | → | 1    |       |
| 2  | → | 8    | 2 ⇒ 8 |
| 3  | → | 27   | 3 ⇒ 7 |
| 4  | → | 64   |       |
| 5  | → | 125  |       |
| 6  | → | 216  |       |
| 7  | → | 343  |       |
| 8  | → | 512  |       |
| 9  | → | 729  |       |
| 10 | → | 1000 |       |

(57)  $x$  ની કિંમત શોધો :  $x^3 \div 11 = 968$ 

(A) 24 (B) 20 (C) 26 (D) 22

Solution :

$$\therefore \frac{x^3}{11} = 968$$

$$\therefore x^3 = \underline{968 \times 11}$$

ગુણકાર  $8 \times 1 = 8$  ઘનમૂળ  $\rightarrow 2$ 

$$\therefore x = \underline{22}$$

$$\therefore (D) 22$$

(58)  $x$  ની કિંમત શોધો :  $x^3 \div 29 = 841$ 

(A) 29 (B) 31 (C) 30 (D) 28

Solution :

$$\therefore \frac{x^3}{29} = 841$$

$$\therefore x^3 = \underline{841 \times 29}$$

ગુણકાર  $9 \times 1 = 9$  ઘનમૂળ  $\rightarrow 9$ 

$$\therefore x = \underline{29}$$

$$\therefore (A) 29$$

(59)  $x$  ની કિંમત શોધો :  $x^3 \div 14 = 1568$ 

(A) 28 (B) 30 (C) 32 (D) 26

Solution :

$$\therefore x^3 = 1568 \times 14 \quad \begin{array}{l} 8 \times 4 \\ \boxed{32} \rightarrow 8 \end{array}$$

$$\therefore (A) 28$$

□ સમજૂતી :

⇒ આધાર સરખા વચ્ચે ગુણકારની નિશાની તો ઘાતનો સરવાળો

$$\therefore a^3 \times a^2 = a^{3+2} = a^5 \times \rightarrow +$$

⇒ આધાર સરખા વચ્ચે ભાગકારની નિશાની તો ઘાતનો બાદબાકી

$$\therefore a^5 \div a^3 = a^{5-3} = a^2 \quad \div \rightarrow -$$

⇒ ઘાતની ઘાતનો ગુણકાર

$$\therefore (a^2)^3 = a^{2 \times 3} = a^6$$

⇒ કોંસ વગરની ઘાત હોય તો ઘાતની ઘાત કરવી.

$$\begin{aligned} \therefore a^{2^3} &= (2 \times 2 \times 2) \\ &= 8 \\ &= a^8 \end{aligned}$$

(60) સાદુંરૂપ આપો :  $324^2 \times 72 \div 18^5 \times 1021$

- (A) 4054 (B) 4064  
(C) 4074 (D) 4084

**Solution :**

$$\begin{aligned} \therefore 324^2 \times 72 \div 18^5 \times 1021 \\ &= \frac{(18^2)^2 \times 18^1 \times 4}{18^5} \times 1021 \\ &= \frac{18^4 \times 18^1 \times 4}{18^5} \times 1021 \\ &= \frac{18^{\cancel{4}} \times 4}{18^{\cancel{4}}} \times 1021 \\ &= 4 \times 1021 \\ &= 4084 \end{aligned}$$

$\therefore$  (D) 4084

(61) સાદુંરૂપ આપો :  $(7^4)^2 \times \frac{1}{343^2} \times 7 + 17 = ?$

- (A) 360 (B) 380  
(C) 340 (D) 320

**Solution :**

$$\begin{aligned} \therefore (7^4)^2 \times \frac{1}{343^2} \times 7 + 17 &= ? \\ &= 7^8 \times \frac{1}{(7^3)^2} \times 7^1 + 17 \\ &= \frac{7^9 \times 1}{7^6} + 17 \\ &= 7^{9-6} + 17 \\ &= 7^3 + 17 \\ &= 343 + 17 \\ &= 360 \end{aligned}$$

$\therefore$  (A) 360

(62) સાદુંરૂપ આપો:  $289^2 \times 68 \div 17^5 \times 1021$

- (A) 4074 (B) 4064  
(C) 4054 (D) 4084

**Solution :**

$$\begin{aligned} \therefore \frac{(17^2)^2 \times 17 \times 4}{17^5} \times 1021 \\ &= \frac{17^{\cancel{4}} \times 4 \times 1021}{17^{\cancel{4}}} \\ &= 4084 \end{aligned}$$

$\therefore$  (D) 4084

(63) સાદુંરૂપ આપો :  $((4^4)^2 \times \frac{1}{64^2})^2 + 10$

- (A) 344 (B) 244  
(C) 366 (D) 266

**Solution :**

$$\begin{aligned} \therefore ((4^4)^2 \times \frac{1}{64^2})^2 + 10 \\ &= (4^8 \times \frac{1}{(4^3)^2})^2 + 10 \\ &= (4^8 \times \frac{1}{4^6})^2 + 10 \\ &= (4^{8-6})^2 + 10 \\ &= (4^2)^2 + 10 \\ &= 16^2 + 10 \\ &= 256 + 10 \\ &= 266 \end{aligned}$$

$\therefore$  (D) 266

(64) સાદુંરૂપ આપો :  $5 \div 5^{-2} + \sqrt{25} - \sqrt{121}$

- (A) 143 (B) 132  
(C) 121 (D) 119

**Solution :**

$$\begin{aligned} \therefore 5 \div 5^{-2} + \sqrt{25} - \sqrt{121} \quad \left/ \quad \begin{array}{l} a^2 = \frac{1}{a^{-2}} \\ \frac{1}{5^{-2}} = 5^2 \text{ થાય} \end{array} \right. \\ &= 5 \times \frac{1}{5^{-2}} + 5 - 11 \\ &= 5 \times 5^2 + 5 - 11 \\ &= 125 + 5 - 11 \\ &= 130 - 11 \\ &= 119 \end{aligned}$$

$\therefore$  (D) 119

(65)  $((6^4)^2 \times \frac{1}{216^2}) \times 6 + 16 = ?$

- (A) 224 (B) 242  
(C) 216 (D) 232

**Solution :**

$$\begin{aligned} \therefore ((6^4)^2 \times \frac{1}{216^2}) \times 6 + 16 &= ? \\ &= (6^8 \times \frac{1}{(6^3)^2}) \times 6^1 + 16 \\ &= \frac{6^9}{6^6} + 16 \\ &= 6^{9-6} + 16 \\ &= 6^3 + 16 \\ &= 216 + 16 \\ &= 232 \end{aligned}$$

$\therefore$  (D) 232



(66) સાદુંરૂપ આપો :  $6 \div 6^{-2} + \sqrt{36} - \sqrt{144}$ 

- (A) 240 (B) 230  
(C) 220 (D) 210

Solution :

$$\begin{aligned}
 &= 6 \div 6^{-2} + \sqrt{36} - \sqrt{144} \\
 &= 6 \times \frac{1}{6^{-2}} + 6 - 12 \\
 &= 6 \times 6^2 + 6 - 12 \\
 &= 6^3 - 6 \\
 &= 216 - 6 \\
 &= 210
 \end{aligned}$$

 $\therefore$  (D) 210(67) સાદુંરૂપ આપો :  $(8^4)^2 \times \frac{1}{512^2} \times 8 + 18 = ?$ 

- (A) 490 (B) 530  
(C) 510 (D) 550

Solution :

$$\begin{aligned}
 \therefore (8^4)^2 \times \frac{1}{512^2} \times 8 + 18 &= ? \\
 &= 8^8 \times \frac{1}{8^6} \times 8^1 + 18 \\
 &= \frac{8^9}{8^6} + 18 \\
 &= 8^{9-6} + 18 \\
 &= 512 + 18 \\
 &= 530
 \end{aligned}$$

 $\therefore$  (B) 530(68) સાદુંરૂપ આપો :  $169^2 \times 52 \div 13^5 \times 1020$ 

- (A) 4050 (B) 4060  
(C) 4070 (D) 4080

Solution :

$$\begin{aligned}
 &= \frac{(13^2)^2 \times 13 \times 4}{13^5} \times 1020 \\
 &= \frac{13^{\cancel{5}}}{13^5} \times 4 \times 1020 \\
 &= 4080
 \end{aligned}$$

 $\therefore$  (D) 4080

## Type # 10

(69) સાદુંરૂપ આપો :  $\frac{23 \times 552 \div 24 + 23}{552 \div 23 - 1} = ?$ 

- (A) 25 (B) 23 (C) 26 (D) 24

Solution :

$$\begin{aligned}
 &= \frac{23 \times \frac{552}{24} + 23}{\frac{552}{23} - 1} \\
 &= \frac{23 \times 23 + 23}{24 - 1} \\
 &= \frac{23 \times 23}{23} + \frac{23}{23} \\
 &= 23 + 1 \\
 &= 24
 \end{aligned}$$

 $\therefore$  (D) 24(70) સાદુંરૂપ આપો :  $\frac{11 \times 132 \div 12 + 11}{108 \div 9 - 1}$ 

- (A) 14 (B) 13 (C) 12 (D) 11

Solution :

$$\begin{aligned}
 &= \frac{11 \times \frac{132}{12} + 11}{\frac{108}{9} - 1} \\
 &= \frac{11 \times 11 + 11}{12 - 1} \\
 &= \frac{11 \times \cancel{11} + \cancel{11}}{\cancel{11}} \\
 &= 11 + 1 \\
 &= 12
 \end{aligned}$$

 $\therefore$  (C) 12(71) સાદુંરૂપ આપો :  $\frac{29 \times 870 \div 30 + 29}{870 \div 29 - 1}$ 

- (A) 32 (B) 31 (C) 29 (D) 30

Solution :

$$\begin{aligned}
 &= \frac{29 \times \frac{870}{30} + 29}{\frac{870}{29} - 1} \\
 &= \frac{29 \times 29 + 29}{29 - 1}
 \end{aligned}$$

$$= \frac{29 \times 29 + 29}{30 - 1}$$

$$= \frac{29 \times \cancel{29}}{\cancel{29}} + \frac{\cancel{29}^1}{\cancel{29}}$$

$$= 29 + 1$$

$$= 30$$

$$\therefore (D) 30$$

(72) સાદુંરૂપ આપો :  $\frac{30 \times 930 \div 31 + 30}{930 \div 30 - 1}$

(A) 33 (B) 31 (C) 34 (D) 32

**Solution :**

$$30 \times \frac{\cancel{930}^{30}}{\cancel{31}} + 30$$

$$= \frac{31}{\frac{\cancel{930}^{30}}{\cancel{30}} - 1}$$

$$= \frac{30 \times 30}{30} + 30$$

$$= \frac{30 \times \cancel{30}}{\cancel{30}} + \frac{\cancel{30}^1}{\cancel{30}}$$

$$= 30 + 1$$

$$= 31$$

$$\therefore (B) 31$$

(73) સાદુંરૂપ આપો :  $\frac{5}{4} + \frac{3}{5} - \frac{9}{10} = ?$

(A) 0.94 (B) 0.96  
(C) 0.97 (D) 0.95

**Solution :**

$$\Rightarrow \text{લ.સ.અ.} = 20$$

$$= \frac{\frac{5}{\cancel{4}}(\cancel{20}^5) + \frac{3}{\cancel{5}}(\cancel{20}^4) - \frac{9}{\cancel{10}}(\cancel{20}^2)}{20}$$

$$= \frac{25 + 12 - 18}{20}$$

$$= \frac{19 \times 5}{20 \times 5} \text{ (અંશ અને છેદમાં 5 વડે ગુણતાં)}$$

$$= \frac{95}{100}$$

$$= 0.95$$

$$\therefore (D) 0.95$$

(74) સાદુંરૂપ આપો :  $\frac{37}{20} + \frac{3}{10} - \frac{5}{4} = ?$

(A) 0.90 (B) 0.91 (C) 0.89 (D) 0.92

**Solution :**

$$\Rightarrow \text{લ.સ.અ.} = 20$$

$$= \frac{\frac{37}{\cancel{20}}(\cancel{20}^1) + \frac{3}{\cancel{10}}(\cancel{20}^2) - \frac{5}{\cancel{4}}(\cancel{20}^5)}{20}$$

$$= \frac{37 + 6 - 25}{20}$$

$$= \frac{43 - 25}{20}$$

$$= \frac{\cancel{18}^9}{\cancel{20}^{10}}$$

$$= 0.9$$

$$\therefore (A) 0.90$$

(75) સાદુંરૂપ આપો :  $\frac{1}{8} + \frac{3}{4} - \frac{5}{8}$

(A) 0.3 (B) 0.45 (C) 0.5 (D) 0.25

**Solution :**

$$= \frac{\frac{1}{\cancel{8}}(\cancel{8}^1) + \frac{3}{\cancel{4}}(\cancel{8}^2) - \frac{5}{\cancel{8}}(\cancel{8}^1)}{8}$$

$$= \frac{1 + 6 - 5}{8}$$

$$= \frac{7 - 5}{8}$$

$$= \frac{\cancel{2}^1}{\cancel{8}^4}$$

$$= 0.25$$

$$\therefore (D) 0.25$$

(76) સાદુંરૂપ આપો :

$$\{2^4 \div 8\}^4 + \sqrt{(135 - \sqrt{196})}$$

(A) 42 (B) 37 (C) 27 (D) 13

**Solution :**

$$= \{2^4 \div 8\}^4 + \sqrt{(135 - \sqrt{196})}$$

$$= \left(\frac{2^4}{2^3}\right)^4 + \sqrt{(135 - 14)}$$

$$\begin{aligned}
 &= (2^{4-3})^4 + \sqrt{121} \\
 &= 2^4 + 11 \\
 &= 16 + 11 \\
 &= 27
 \end{aligned}$$

∴ (C) 27

(77) સાદુરૂપ આપો :  $486 \div \left[ \frac{3}{4} \left\{ \frac{16}{15} \right\} - \frac{2}{3} \right]$

(A) 3654 (B) 3645 (C) 3456 (D) 3465

**Solution :**

$$\begin{aligned}
 &= 486 \div \left( \frac{\cancel{4}}{\cancel{4}} \left[ \frac{\cancel{4}}{\cancel{15}} \right] - \frac{2}{3} \right) \\
 &= 486 \div \left( \frac{4}{5} - \frac{2}{3} \right) \\
 &= 486 \div \left( \frac{12-10}{15} \right) \\
 &= 486 \div \left( \frac{2}{15} \right) \\
 &= \cancel{486}^{\cancel{243}} \times \frac{15}{\cancel{2}} \\
 &= 243 \times 15 \\
 &= 3645 \\
 &\therefore \text{(B) 3645}
 \end{aligned}$$

(78) સાદુરૂપ આપો :  $468 \div \left[ \frac{3}{4} \left\{ \frac{16}{15} \right\} - \frac{2}{3} \right]$

(A) 3510 (B) 3610 (C) 3410 (D) 3710

**Solution :**

$$\begin{aligned}
 &= 468 \div \left( \frac{\cancel{4}}{\cancel{4}} \left[ \frac{\cancel{4}}{\cancel{15}} \right] - \frac{2}{3} \right) \\
 &= 468 \div \left( \frac{4}{5} - \frac{2}{3} \right) \\
 &= 468 \div \left( \frac{12-10}{15} \right) \\
 &= 468 \div \frac{2}{15} \\
 &= \cancel{468}^{\cancel{234}} \times \frac{15}{\cancel{2}} \\
 &= 234 \times 15 \\
 &= 3510 \\
 &\therefore \text{(A) 3510}
 \end{aligned}$$

(79) સાદુરૂપ આપો :

$$[2(54 \times 17 \times 54 \times 17) \div (2(\sqrt{289} \div 2))^2]$$

(A) 2601

(B) 2916

(C) 2704

(D) 2401

**Solution :**

$$\begin{aligned}
 &= 2(54 \times 17 \times 54 \times 17) \div \left( 2 \left( \sqrt{\frac{289}{2}} \right) \right)^2 \\
 &= 2(54 \times 17 \times 54 \times 17) \div \left( 2 \left( \frac{17}{\sqrt{2}} \right) \right)^2 \\
 &= 2(54 \times 17 \times 54 \times 17) \div \left( \cancel{4}^2 \times \frac{17 \times 17}{\cancel{2}} \right) \\
 &= \frac{\cancel{2} (54 \times \cancel{17} \times 54 \times \cancel{17})}{\cancel{2} \times \cancel{17} \times \cancel{17}} \\
 &= 54 \times 54 \\
 &= 2916 \\
 &\therefore \text{(B) 2916}
 \end{aligned}$$

(80) સાદુરૂપ આપો :

$$(21 \times 8 \times 21 \times 16) \div (2(\sqrt{64} \div 2))^2$$

(A) 441

(B) 484

(C) 144

(D) 324

**Solution :**

$$\begin{aligned}
 &= (21 \times 8 \times 21 \times 16) \div (2(\sqrt{64} \div 2))^2 \\
 &= (21 \times 8 \times 21 \times 16) \div \left( 2 \left( \sqrt{\frac{64}{2}} \right) \right)^2 \\
 &= (21 \times 8 \times 21 \times 16) \div (2(\sqrt{32}))^2 \\
 &= (21 \times 8 \times 21 \times 16) \div (4 \times 32) \\
 &= \frac{21 \times \cancel{8}^2 \times 21 \times \cancel{16}}{\cancel{4} \times \frac{\cancel{32}}{2}} \\
 &= \frac{21 \times 21 \times \cancel{2}}{\cancel{2}} \\
 &= 21 \times 21 \\
 &= 441 \\
 &\therefore \text{(A) 441}
 \end{aligned}$$

(81) સાદુરૂપ આપો :

$$(38 \times 33 \times 38 \times 33) \div (2(\sqrt{1089} \div 2)^2)$$

(A) 1444 (B) 5444 (C) 1222 (D) 2444

Solution :

$$= 38 \times 33 \times 38 \times 33 \div \left( 2 \left( \sqrt{\frac{1089}{2}} \right)^2 \right)$$

$$= 38 \times 33 \times 38 \times 33 \div \left( 2 \left( \frac{33}{\sqrt{2}} \right)^2 \right)$$

$$= 38 \times 33 \times 38 \times 33 \div \left( \cancel{2} \left( \frac{33 \times 33}{\cancel{2}} \right) \right)$$

$$= \frac{38 \times \cancel{33} \times 38 \times \cancel{33}}{\cancel{33} \times \cancel{33}}$$

$$= 38 \times 38$$

$$= 1444$$

$$\therefore (A) 1444$$

(82)  $\left\{ \frac{12}{9-6} \right\} + 8 - \left( \frac{16}{4} \right) - (3 - 5 - 4)$  ની કિંમત શોધો.

(A) 14 (B) 35 (C) 42 (D) 24

Solution :

$$= \left\{ \frac{12}{9-6} \right\} + 8 - \left( \frac{16}{4} \right) - (3 - 5 - 4)$$

$$= \left\{ \frac{12}{3} \right\} + 8 - (4) - (3 - 9)$$

$$= \left\{ \frac{12}{3} \right\} + 8 - (4) - (-6)$$

$$= \left\{ \frac{12}{3} \right\} + 8 - 4 + 6$$

$$= 4 + 8 - 4 + 6$$

$$= 14$$

$$\therefore (A) 14$$

(83)  $3\frac{1}{2} \div \left\{ 1\frac{1}{4} - \left( \frac{1}{2} + \frac{1}{3} \right) \right\} + \frac{7}{5}$  નું સાદુરૂપ આપો.(A) 5 (B) 7 (C)  $9\frac{4}{5}$  (D)  $8\frac{5}{6}$ 

Solution :

$$= \left[ 3\frac{1}{2} \div \left\{ 1\frac{1}{4} - \left( \frac{1}{2} + \frac{1}{3} \right) \right\} + \frac{7}{5} \right]$$

$$= \left[ \frac{7}{2} \div \left\{ \frac{5}{4} - \frac{5}{6} \right\} + \frac{7}{5} \right]$$

$$= \left[ \frac{7}{2} \div \left\{ \frac{30-20}{24} \right\} + \frac{7}{5} \right]$$

$$= \left[ \frac{7}{2} \div \left\{ \frac{10}{24} \right\} + \frac{7}{5} \right]$$

$$= \left[ \frac{7}{\cancel{2}} \times \frac{\cancel{24}^6}{\cancel{10}_5} + \frac{7}{5} \right]$$

$$= \frac{42}{5} + \frac{7}{5}$$

$$= \frac{42+7}{5}$$

$$= \frac{49}{5}$$

$$= 9\frac{4}{5}$$

$$\therefore (C) 9\frac{4}{5}$$

## Type # 11

□ સમજૂતી :

⇒ દશાંશ આપેલ હોય તેવી સંખ્યાનું વર્ગમૂળ કાઢતા આપેલ દશાંશના અડધા વર્ગમૂળની બહાર આવે.

$$\Rightarrow \sqrt{3136} = 56$$

$$\Rightarrow \sqrt{31.36} = 5.6$$

$$\Rightarrow \sqrt{0.3136} = 0.56$$

$$\Rightarrow \sqrt{0.003136} = 0.056$$

$$\Rightarrow \sqrt{0.00003136} = 0.0056$$

(84)  $x$  ની કિંમત શોધો :  $\frac{2}{5}(x) + \frac{3}{10}(x) - \frac{3}{5}(x) = 531$ 

(A) 5210 (B) 5410 (C) 5310 (D) 5510

Solution :

$$\Rightarrow \frac{2}{5}x + \frac{3}{10}x - \frac{3}{5}x = 531$$

$$\frac{2x}{5} + \frac{3x}{10} - \frac{3x}{5} = 531$$

$$\frac{4x + 3x - 6x}{10} = 531$$

$$\frac{x}{10} = 531$$

$$x = 531 \times 10$$

$$x = 5310$$

$$\therefore (C) 5310$$

(85)  $x$  ની કિંમત શોધો :

$$\frac{2}{5}(x) + \frac{3}{10}(x) - \frac{3}{5}(x) = 521$$

- (A) 5310 (B) 5110  
(C) 5210 (D) 5410

Solution :

$$\Rightarrow \frac{2}{5}x + \frac{3}{10}x - \frac{3}{5}x = 521$$

$$\frac{2x}{5} + \frac{3x}{10} - \frac{3x}{5} = 521$$

$$\frac{4x + 3x - 6x}{10} = 521$$

$$\frac{x}{10} = 521$$

$$x = 521 \times 10$$

$$x = 5210$$

∴ (C) 5210

(86)  $x$  ની કિંમત શોધો :

$$15.73 + 13.25 + 16.73 - 28.64 = 3x$$

- (A) 5.69 (B) 5.49  
(C) 5.59 (D) 5.79

Solution :

$$\Rightarrow 15.73 + 13.25 + 16.73 - 28.64 = 3x$$

$$45.71 - 28.64 = 3x$$

$$\frac{17.07}{3} = x$$

$$x = 5.69$$

∴ (A) 5.69

(87) સાદુરૂપ આપો :  $\sqrt{20.25} + \sqrt{75.69} + 8.7$ 

- (A) 21.5 (B) 21.7  
(C) 21.9 (D) 21.3

Solution :

|                       |                        |
|-----------------------|------------------------|
| 45                    | 87                     |
| 4 $\overline{)20.25}$ | 8 $\overline{)75.69}$  |
| +4 $\underline{16}$   | +8 $\underline{64}$    |
| 85 $\underline{425}$  | 167 $\underline{1169}$ |
| $\underline{425}$     | $\underline{1169}$     |
| 000                   | 0000                   |

$$\Rightarrow 4.5 + 8.7 + 8.7 = 21.9$$

∴ (C) 21.9

(88) સાદુરૂપ આપો :  $\sqrt{21.16} + \sqrt{79.21} + 8.9$ 

- (A) 22.6 (B) 22.4  
(C) 22.2 (D) 22.8

Solution :

|                      |                        |
|----------------------|------------------------|
| 46                   | 89                     |
| 4 $\overline{)2116}$ | 8 $\overline{)7921}$   |
| +4 $\underline{16}$  | +8 $\underline{64}$    |
| 86 $\underline{516}$ | 169 $\underline{1521}$ |
| $\underline{516}$    | $\underline{1521}$     |
| 000                  | 0000                   |

$$\therefore 4.6 + 8.9 + 8.9 = 22.4$$

∴ (B) 22.4

(89) સાદુરૂપ આપો :  $(13)^3 + (-8)^3 + (-5)^3$ 

- (A)  $-40 \times 40$  (B)  $-39 \times 40$   
(C)  $39 \times 40$  (D)  $41 \times 38$

Solution :

⇒ જ્યારે એકી ઘાત હોય, આધાર માઈનસ (-) હોય તો તેનું સાદુરૂપ માઈનસ (-) આવે.

$$\begin{aligned} \Rightarrow (13)^3 + (-8)^3 + (-5)^3 \\ = 2197 + (-512) + (-125) \\ = 2197 - 637 \\ = 1560 \\ = 39 \times 40 \end{aligned}$$

∴ (C)  $39 \times 40$

(90)  $\sqrt{16.81} + \sqrt{67.24} \div 2$ 

- (A) 8.4 (B) 8  
(C) 8.2 (D) 8.1

Solution :

|                       |                        |
|-----------------------|------------------------|
| 41                    | 82                     |
| 4 $\overline{)1681}$  | 8 $\overline{)6724}$   |
| +4 $\underline{16}$   | +8 $\underline{64}$    |
| 81 $\underline{0081}$ | 162 $\underline{0324}$ |
| $\underline{81}$      | $\underline{324}$      |
| 00                    | 000                    |

$$\Rightarrow 4.1 + (8.2 \div 2) = 4.1 + 4.1 = 8.2$$

∴ (C) 8.2



(91) પ્રશ્નાર્થની જગ્યાએ શું આવશે ?

$$12.34 + 23.45 + 34.56 - 45.67 = 2 \times ?$$

- (A) 13.34 (B) 14.34  
(C) 12.34 (D) 11.34

**Solution :**

$$\Rightarrow 12.34 + 23.45 + 34.56 - 45.67 = 2 \times ?$$

$$70.35 - 45.67 = 2 \times ?$$

$$24.68 = 2x$$

$$x = \frac{24.68}{2}$$

$$x = 12.34$$

$$\therefore (C) 12.34$$

(92)  $x$  ની કિંમત શોધો :

$$15.37 + 13.52 + 16.37 - 28.46 = 3x$$

- (A) 5.7 (B) 5.5 (C) 5.4 (D) 5.6

**Solution :**

$$\Rightarrow 15.37 + 13.52 + 16.37 - 28.46 = 3x$$

$$45.26 - 28.46 = 3x$$

$$16.8 = 3x$$

$$x = \frac{16.8}{3}$$

$$x = 5.6$$

$$\therefore (D) 5.6$$

(93)  $x$  ની કિંમત શોધો :

$$\frac{2}{5}(x) + \frac{3}{10}(x) - \frac{3}{5}(x) = 439$$

- (A) 4390 (B) 4590  
(C) 4290 (D) 4490

**Solution :**

$$\Rightarrow \frac{4x + 3x - 6x}{10} = 439$$

$$\frac{x}{10} = 439$$

$$x = 439 \times 10$$

$$x = 4390$$

$$\therefore (A) 4390$$

(94) સાદુંરૂપ આપો :

$$143 + 14.3 + 1.43 + 0.143 + 0.0143 = ?$$

- (A) 158.8783 (B) 158.8863  
(C) 158.8683 (D) 158.8873

**Solution :**

$$\begin{array}{r} 143.0 \\ + 14.3 \\ + 1.43 \\ + 0.143 \\ + 0.0143 \\ \hline 158.8873 \end{array}$$

$$\therefore (D) 158.8873$$

(95)  $0.1 + 0.01 + 0.001 + 0.0001 + 0.00001 = ?$

- (A) 0.10101 (B) 0.11111  
(C) 0.00001 (D) 0.01010

**Solution :**

$$\begin{array}{r} 0.1 \\ + 0.01 \\ + 0.001 \\ + 0.0001 \\ + 0.00001 \\ \hline 0.11111 \end{array}$$

$$\therefore (B) 0.11111$$

(96)  $0.1 + 0.11 + 0.111 + 0.1111 + 0.11111 = ?$

- (A) 0.54321 (B) 0.54325  
(C) 0.54252 (D) 0.54378

**Solution :**

$$\begin{array}{r} 0.1 \\ + 0.11 \\ + 0.111 \\ + 0.1111 \\ + 0.11111 \\ \hline 0.54321 \end{array}$$

$\Rightarrow$  (આપેલી સંખ્યાનો અંતિમ અંકથી એક એક ને ઉમેરીને સરવાળો કરવો.)

$$\therefore (A) 0.54321$$

(97)  $0.3 + 0.33 + 0.333 + 0.3333 + 0.33333 = ?$

- (A) 15.2963 (B) 16.2963  
(C) 1.62963 (D) 1.52963

**Solution :**

$$\begin{array}{r} 0.3 \\ + 0.33 \\ + 0.333 \\ + 0.3333 \\ + 0.33333 \\ \hline 1.62963 \end{array}$$

$$\therefore (C) 1.62963$$

(98)  $0.5 + 0.55 + 0.555 = ?$ 

- (A) 1.605 (B) 1.505  
(C) 0.1605 (D) 0.1505

Solution :

$$\begin{array}{r} 0.5 \\ + 0.55 \\ + 0.555 \\ \hline 1.605 \end{array}$$

 $\therefore$  (A) 1.605

## Type # 12

□ સમજૂતી :

|             |    |         |         |        |         |
|-------------|----|---------|---------|--------|---------|
| B           | O  | D       | M       | A      | S       |
| ↓           | ↓  | ↓       | ↓       | ↓      | ↓       |
| 1.          | 2. | 3.      | 4.      | 5.     | 6.      |
| બ્રેકેટ     | OF | ભાગાકાર | ગુણાકાર | સરવાળો | બાદબાકી |
| → 0, {}, [] |    |         |         |        |         |

⇒ સાદુંરૂપ આપતી વખતે નીચેના ક્રમો ધ્યાને રાખી અનુક્રમે સાદુંરૂપ આપવું.

1.  $13 - 5$  (બારની નિશાની)
2. ( )
3. { }
4. [ ]
5. of (નો, ની, નું, ના)
6. ભાગાકાર
7. ગુણાકાર
8. સરવાળો
9. બાદબાકી

(99) સાદુંરૂપ આપો :  $48 - [15 + \{38 - (28 - 7)\}]$ 

- (A) 19 (B) 18  
(C) 16 (D) 17

Solution :

$$\begin{aligned} &= 48 - [15 + \{38 - (28 - 7)\}] \\ &= 48 - [15 + \{38 - 21\}] \\ &= 48 - [15 + \{17\}] \\ &= 48 - 32 \\ &= 16 \end{aligned}$$

 $\therefore$  (C) 16

(100) સાદુંરૂપ આપો :

$$79 - [23 - \{84 \div 7 - (16 - 4) \div 4\}]$$

- (A) 61 (B) 63 (C) 67 (D) 65

Solution :

$$\begin{aligned} &= 79 - [23 - \{84 \div 7 - (16 - 4) \div 4\}] \\ &= 79 - [23 - \{12 - 3\}] \\ &= 79 - [23 - 9] \\ &= 79 - 14 \\ &= 65 \end{aligned}$$

 $\therefore$  (D) 65(101) સાદુંરૂપ આપો :  $49 - [16 + \{39 - (29 - 7)\}]$ 

- (A) 17 (B) 19 (C) 18 (D) 16

Solution :

$$\begin{aligned} &= 49 - [16 + \{39 - (29 - 7)\}] \\ &= 49 - [16 + \{39 - 22\}] \\ &= 49 - [16 + 17] \\ &= 49 - 33 \\ &= 16 \end{aligned}$$

 $\therefore$  (D) 16

(102) સાદુંરૂપ આપો :

$$250 - [90 - \{120 \div 10 - (30 - 60 \div 3) \div 10\}]$$

- (A) 143 (B) 171 (C) 134 (D) 166

Solution :

$$\begin{aligned} &250 - [90 - \{120 \div 10 - (30 - 60 \div 3) \div 10\}] \\ &= 250 - [90 - \{120 \div 10 - (30 - 20) \div 10\}] \\ &= 250 - [90 - \{12 - 1\}] \\ &= 250 - [90 - 11] \\ &= 250 - 79 \\ &= 171 \end{aligned}$$

 $\therefore$  (B) 171

(103) સાદુંરૂપ આપો :

$$120 - [100 - \{81 \div 9 - (65 - 100 \div 2) \div 15\}]$$

- (A) 36 (B) 32 (C) 28 (D) 34

Solution :

$$\begin{aligned} &120 - [100 - \{81 \div 9 - (65 - 100 \div 2) \div 15\}] \\ &= 120 - [100 - \{9 - (65 - 50) \div 15\}] \end{aligned}$$

$$\begin{aligned}
 &= 120 - [100 - \{9 - (15 \div 15)\}] \\
 &= 120 - [100 - \{9 - 1\}] \\
 &= 120 - [100 - 8] \\
 &= 120 - 92 \\
 &= 28 \\
 &\therefore \text{(C) } 28
 \end{aligned}$$

(104) સાર્વજનિક આપો :

$$48 \div [35 - \{30 - (31 - 26 \div 2 \times 4)\}] = ?$$

(A) -2      (B) -3      (C) -4      (D) -8

Solution :

$$\begin{aligned}
 \Rightarrow 48 \div [35 - \{30 - (31 - \cancel{26} \div \cancel{2} \times 4)\}] \\
 &= 48 \div [35 - \{30 - (31 - 13 \times 4)\}] \\
 &= 48 \div [35 - \{30 - (31 - 52)\}] \\
 &= 48 \div [35 - \{30 + 21\}] \\
 &= 48 \div [35 - 51] \\
 &= \cancel{48} \div [-\cancel{16}] \\
 &= -3 \\
 &\therefore \text{(B) } -3
 \end{aligned}$$

(105)  $72 - [20 - \{10 - (6 - \overline{6-4})\}]$  ની કિંમત શોધો.

(A) 58      (B) 48      (C) 56      (D) 44

Solution :

$$\begin{aligned}
 \Rightarrow 72 - [20 - \{10 - (6 - \overline{6-4})\}] \\
 &= 72 - [20 - \{10 - (6 - 2)\}] \\
 &= 72 - [20 - \{10 - 4\}] \\
 &= 72 - [20 - \{+6\}] \\
 &= 72 - 14 \\
 &= 58 \\
 &\therefore \text{(A) } 58
 \end{aligned}$$

(106)  $48 + [25 - \{20 - (11 - 16 \div 2 \times 4)\}]$

(A) 40      (B) 32      (C) 62      (D) 58

Solution :

$$\begin{aligned}
 \Rightarrow 48 + [25 - \{20 - (11 - \cancel{16} \div \cancel{2} \times 4)\}] \\
 &= 48 + [25 - \{20 - (11 - (8 \times 4))\}] \\
 &= 48 + [25 - \{20 - (11 - 32)\}] \\
 &= 48 + [25 - \{20 + 21\}] \\
 &= 48 + [25 - 41] \\
 &= 48 + [-16]
 \end{aligned}$$

$$\begin{aligned}
 &= 48 - 16 \\
 &= 32
 \end{aligned}$$

$$\therefore \text{(B) } 32$$

(107)  $72 \div [38 - \{30 - (31 - 60 \div 4 \times 5)\}] = ?$

(A) -2      (B) -3      (C) -4      (D) -8

Solution :

$$\begin{aligned}
 \Rightarrow 72 \div [38 - \{30 - (31 - (\cancel{60} \div \cancel{4} \times 5))\}] \\
 &= 72 \div [38 - \{30 - (31 - (15 \times 5))\}] \\
 &= 72 \div [38 - \{30 - (31 - 75)\}] \\
 &= 72 \div [38 - \{30 - (-44)\}] \\
 &= 72 \div [38 - \{30 + 44\}] \\
 &= 72 \div [38 - 74] \\
 &= \cancel{72} \div [-\cancel{36}] \\
 &= -2 \\
 &\therefore \text{(A) } -2
 \end{aligned}$$

(108)  $113 + 11.3 + 1.13 + 0.113 + 0.0113 = ?$

(A) 125.5643      (B) 125.5453  
(C) 125.5553      (D) 125.5543

Solution :

$$\begin{array}{r}
 113.0 \\
 + 11.3 \\
 + 1.13 \\
 + 0.113 \\
 + 0.0113 \\
 \hline
 125.5543 \\
 \therefore \text{(D) } 125.5543
 \end{array}$$

(109)  $256 + 25.6 + 2.56 + 0.256 + 0.0256 = ?$

(A) 284.6536      (B) 284.4666  
(C) 284.4766      (D) 284.4416

Solution :

$$\begin{array}{r}
 256.0 \\
 + 25.6 \\
 + 2.56 \\
 + 0.256 \\
 + 0.0256 \\
 \hline
 284.4416 \\
 \therefore \text{(D) } 284.4416
 \end{array}$$

(110)  $110 \div [53 - \{4^3 - \{37 - 136 \div \overline{17 \times 2}\}\}] = ?$

(A) 11 (B) 22 (C) 5 (D) 10

**Solution :**

$$\begin{aligned} \Rightarrow 110 \div [53 - \{4^3 - \{37 - 136 \div \overline{17 \times 2}\}\}] &= ? \\ &= 110 \div [53 - \{4^3 - \{37 - 136 \div 17 \times 2\}\}] \\ &= 110 \div [53 - \{4^3 - (37 - \frac{136}{34})\}] \\ &= 110 \div [53 - \{4^3 - \{37 - 4\}\}] \\ &= 110 \div [53 - \{4^3 - \{33\}\}] \\ &= 110 \div [53 - \{64 - 33\}] \\ &= 110 \div [53 - 31] \\ &= \frac{110}{22} \\ &= 5 \end{aligned}$$

$\therefore$  (C) 5

(111)  $[60 \div \{3^3 \times 2 - (29 - \overline{26 - 31})\}] = ?$

(A) 5 (B) 10 (C) 6 (D) 3

**Solution :**

$$\begin{aligned} \Rightarrow [60 \div \{3^3 \times 2 - (29 - \overline{26 - 31})\}] &= ? \\ &= [60 \div \{3^3 \times 2 - (29 - (-5))\}] \\ &= [60 \div \{3^3 \times 2 - (29 + 5)\}] \\ &= [60 \div \{27 \times 2 - 34\}] \\ &= [60 \div \{54 - 34\}] \\ &= \left[ \frac{60}{20} \right] \\ &= 3 \end{aligned}$$

$\therefore$  (D) 3

(112)  $\frac{10 - [5 - (5 + 10)] \div 5}{(-5 + 10) - (5 - 10) \div 5} = ?$

(A) 1 (B) 2 (C) 5 (D) 10

**Solution :**

$$\begin{aligned} \Rightarrow \frac{10 - [5 - (5 + 10)] \div 5}{(-5 + 10) - (5 - 10) \div 5} &= ? \\ &= \frac{10 - [5 - 15] \div 5}{5 - (-5) \div 5} \\ &= \frac{10 - \left[ \frac{2}{-10} \right] \div 5}{5 + (5 \div 5)} \\ &= \frac{10 - (-2)}{5 + 1} \end{aligned}$$

$$= \frac{10 + 2}{5 + 1}$$

$$= \frac{12}{6}$$

$$= 2$$

$\therefore$  (B) 2

(113)  $60 \div 5 \times (16 - 8 \div 2) \div 3 = ?$

(A) 48 (B) 16 (C) 1 (D) 32

**Solution :**

$$\begin{aligned} \Rightarrow 60 \div 5 \times (16 - 8 \div 2) \div 3 &= ? \\ &= 60 \div 5 \times (16 - 4) \div 3 \\ &= \frac{12}{60 \div 5} \times \frac{4}{12 \div 3} \quad (\times \div \text{સાથે થઈ શકે.}) \\ &= 12 \times 4 \\ &= 48 \end{aligned}$$

$\therefore$  (A) 48

(114)  $15 \div 3 \text{ of } 2 \times 2 + 9 \div 18 \text{ of } 2 \times 4 - 4 \div 8 \times 2$   
ની કિંમત શોધો.

(A) 8 (B) 5 (C)  $2\frac{3}{4}$  (D)  $7\frac{3}{4}$

**Solution :**

$$\begin{aligned} \Rightarrow 15 \div 3 \text{ of } 2 \times 2 + 9 \div 18 \text{ of } 2 \times 4 - 4 \div 8 \times 2 &= ? \\ &= 15 \div 6 \times 2 + 9 \div 36 \times 4 - \frac{4}{8} \times 2 \\ &= \frac{15 \times 2}{6} + \frac{1}{4} \times 4 - 1 \\ &= 5 + 1 - 1 \\ &= 5 \end{aligned}$$

$\therefore$  (B) 5

(115)  $20 \div 5 \text{ of } 4 \times 8 + 6 \div 24 \text{ of } 3 \times 48 - 6 \div 18 \times 9$   
નું સાદુરૂપ આપો.

(A) 9 (B) 6 (C) 4 (D) 1.125

**Solution :**

$$\begin{aligned} \Rightarrow 20 \div 5 \text{ of } 4 \times 8 + 6 \div 24 \text{ of } 3 \times 48 - 6 \div 18 \times 9 &= ? \\ &= 1 \times 8 + 6 \div 72 \times 48 - 3 \\ &= 8 + \frac{6 \times 48}{72} - 3 \\ &= 5 + 4 \\ &= 9 \end{aligned}$$

$\therefore$  (A) 9

## Type # 13

□ સમજૂતી :

◆  $0.\overline{3}$  ને અપૂર્ણકમાં દર્શાવો.

$$\therefore 0.\overline{3} = \frac{3}{9} = \frac{1}{3}$$

⇒ જેટલા (-) હોય તેટલા 9 લખવાના

$$\therefore 0.\overline{1} = \frac{1}{9}$$

$$\therefore 0.\overline{12} = \frac{12}{99}$$

$$\therefore 0.\overline{123} = \frac{123}{999}$$

$$\therefore 0.12\overline{3} \rightarrow \frac{123-12}{900} = \frac{111}{900}$$

- સંખ્યા લખી જેમાં બાર કરેલ ન હોય ત બાદ કરો.
- જેટલા બાર તેટલા 9 લખો (છેદમાં)
- જેટલા પર બાર ન હોય તેટલા શૂન્ય લગાવો.

(116)  $x$  ની અપૂર્ણકમાં કિંમત શોધો. :  $x = 0.3444444 \dots$

(A)  $\frac{31}{990}$  (B)  $\frac{35}{99}$  (C)  $\frac{31}{90}$  (D)  $\frac{13}{99}$

Solution :

$$\Rightarrow x = 0.3\overline{4} = \frac{34-3}{90} = \frac{31}{90}$$

$$\therefore (C) \frac{31}{90}$$

(117)  $x$  ની અપૂર્ણકમાં કિંમત શોધો. :  $x = 0.5808080 \dots$

(A)  $\frac{572}{990}$  (B)  $\frac{574}{990}$  (C)  $\frac{573}{990}$  (D)  $\frac{575}{990}$

Solution :

$$\Rightarrow x = 0.5\overline{80} = \frac{580-5}{990} = \frac{575}{990}$$

$$\therefore (D) \frac{575}{990}$$

(118)  $x$  ની કિંમત અપૂર્ણકમાં શોધો. :  $x = 0.3566666 \dots$

(A)  $\frac{324}{900}$  (B)  $\frac{321}{990}$  (C)  $\frac{321}{99}$  (D)  $\frac{321}{900}$

Solution :

$$\Rightarrow x = 0.35\overline{6} = \frac{356-35}{900} = \frac{321}{900}$$

$$\therefore (D) \frac{321}{900}$$

(119)  $x$  ની કિંમત અપૂર્ણકમાં શોધો. :  $x = 0.474747 \dots$

(A)  $\frac{47}{99}$  (B)  $\frac{48}{99}$  (C)  $\frac{46}{90}$  (D)  $\frac{48}{90}$

Solution :

$$\Rightarrow x = 0.4\overline{7} = \frac{47-0}{99} = \frac{47}{99}$$

$$\therefore (A) \frac{47}{99}$$

(120)  $x$  ની કિંમત અપૂર્ણકમાં શોધો. :  $x = 0.8422222 \dots$

(A)  $\frac{741}{900}$  (B)  $\frac{758}{990}$   
(C)  $\frac{758}{99}$  (D)  $\frac{758}{900}$

Solution :

$$\Rightarrow x = 0.84\overline{2} = \frac{842-84}{900} = \frac{758}{900}$$

$$\therefore (D) \frac{758}{900}$$

(121) જો  $0.234343434 \dots$  ને  $\frac{A}{B}$  ના સ્વરૂપમાં દર્શાવવામાં આવે તો  $B - A$  ની કિંમત શું થાય ?

(A) 379 (B) 376  
(C) 378 (D) 377

Solution :

$$\Rightarrow 0.23\overline{4} = \frac{234-2}{990} = \frac{232}{990} = \frac{116}{495}$$

$$116 = A \quad 495 = B$$

$$\therefore B - A = 495 - 116 = 379$$

$$\therefore (A) 379$$

(122)  $1.\overline{6}$  નું સરળ રૂપ લખો.

(A)  $\frac{8}{5}$  (B)  $\frac{5}{3}$   
(C)  $\frac{833}{500}$  (D)  $\frac{16}{33}$

Solution :

$$\Rightarrow 1 + 0.\overline{6} = 1 + \frac{6}{10} = 1 + \frac{2}{5} = \frac{5+2}{5} = \frac{7}{5}$$

$$\therefore (B) \frac{7}{5}$$

(123)  $0.5\overline{87}$  ને સરળ રૂપમાં લખો.

(A)  $\frac{93}{167}$  (B)  $\frac{97}{165}$   
(C)  $\frac{95}{167}$  (D)  $\frac{91}{165}$

Solution :

$$\Rightarrow \frac{587-5}{990} = \frac{582}{990} = \frac{194}{330} = \frac{97}{165}$$

$$\therefore (B) \frac{97}{165}$$



(124) સાદુરૂપ આપો :  $(0.\overline{39} + 0.\overline{53} - 0.\overline{27})$ (A)  $0.\overline{65}$  (B)  $0.\overline{63}$  (C)  $0.\overline{62}$  (D)  $0.\overline{66}$ 

Solution :

$$\begin{array}{r} 0.39 \\ + 0.53 \\ \hline 0.92 \\ - 0.27 \\ \hline 0.65 \end{array}$$

 $\therefore$  (A)  $0.\overline{65}$ (125) સાદુરૂપ આપો :  $(0.\overline{23} + 0.\overline{63} + 0.\overline{15})$ (A)  $2\frac{9}{99}$  (B)  $4\frac{4}{9}$  (C)  $3\frac{3}{99}$  (D)  $1\frac{2}{99}$ 

Solution :

 $\Rightarrow$  જ્યારે છેદ સમાન હોય ત્યારે અંશનો સિધો સરવાળો થાય.

$$\Rightarrow \frac{23}{99} + \frac{63}{99} + \frac{15}{99} = \frac{101}{99} = 1\frac{2}{99}$$

 $\therefore$  (D)  $1\frac{2}{99}$ 

## Type # 14

□ સમજૂતી :

(અંતિમ પદની સંખ્યાને દર્શાવ્યા મુજબ લખો)

- જેટલા સરવાળાની નિશાની હોય તે લખો. ત્યારબાદ બંને કિંમતનો સરવાળો કરો.

$$1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{3}}}}$$

1   3   4   7   11   18

અંશ   છેદ

$\frac{18}{11}$

(126)  $1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{4}}}$  = ?(A)  $\frac{21}{17}$  (B)  $\frac{17}{13}$  (C)  $\frac{22}{13}$  (D)  $\frac{23}{14}$ 

Solution :

$$\begin{array}{r} 4 \quad 5 \quad 9 \quad 14 \quad 23 \\ \hline 23 \\ 14 \end{array}$$

$\therefore$  (D)  $\frac{23}{14}$

$$(127) 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{2}}}}$$

(A)  $\frac{21}{13}$ (B)  $\frac{17}{2}$ (C)  $\frac{34}{21}$ (D)  $\frac{8}{5}$ 

Solution :

$$\begin{array}{r} + \quad + \quad + \quad + \quad + \\ 2 \quad 3 \quad 5 \quad 8 \quad 13 \quad 21 \quad 34 = \frac{34}{21} \\ \uparrow \quad \uparrow \\ \text{છેદ} \quad \text{અંશ} \end{array}$$

 $\therefore$  (C)  $\frac{34}{21}$ (128) જો  $x = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{2}}}$  હોય, તો  $2x + \frac{11}{4}$  ની

કિંમત શોધો.

(A) 6

(B)  $\frac{13}{8}$ 

(C) 4

(D)  $\frac{7}{8}$ 

Solution :

$$\begin{array}{r} + \quad + \quad + \quad + \\ 1 \quad 2 \quad 3 \quad 5 \quad 8 \quad 13 \\ \uparrow \quad \uparrow \\ \text{છેદ} \quad \text{અંશ} \end{array}$$

$$x = \frac{13}{8}$$

$$= 2x + \frac{11}{4} = 2\left(\frac{13}{8}\right) + \frac{11}{4} = \frac{13+11}{4} = \frac{24}{4} = 6$$

 $\therefore$  (A) 6

$$(129) 1 - \frac{1}{1 - \frac{1}{1 - \frac{1}{5}}}$$

(A) 5

(B) 7

(C) 9

(D) 13

Solution :

$$\begin{array}{r} - \quad - \quad - \\ 1 \quad 5 \quad 4 \quad -1 \quad -5 \\ \uparrow \quad \uparrow \\ \text{છેદ} \quad \text{અંશ} \end{array}$$

$$= \frac{-5}{-1} = 5$$

 $\therefore$  (A) 5

$$(130) 1 - \frac{1}{1 - \frac{1}{1 - \frac{1}{1 - \frac{1}{4}}}}$$

(A)  $\frac{2}{5}$  (B)  $\frac{3}{4}$  (C)  $\frac{5}{7}$  (D)  $\frac{7}{3}$

**Solution :**

$$1 - \frac{1}{1 - \frac{1}{1 - \frac{1}{1 - \frac{1}{4}}}} = \frac{3}{4}$$

↑    ↑  
છેદ    અંશ

∴ (B)  $\frac{3}{4}$

$$(131) 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{3}}}}$$

(A)  $\frac{1}{2}$  (B)  $\frac{11}{7}$  (C)  $\frac{3}{4}$  (D)  $\frac{9}{4}$

**Solution :**

$$1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{3}}}} = \frac{9}{4}$$

↑    ↑  
છેદ    અંશ

∴ (D)  $\frac{9}{4}$

### Type # 15

(132) નીચેના અપૂર્ણાંકમાંથી સૌથી મોટો અપૂર્ણાંક કયો ?

$$\frac{1}{2}, \frac{5}{7}, \frac{3}{4}, \frac{6}{7}$$

(A)  $\frac{1}{2}$  (B)  $\frac{5}{7}$  (C)  $\frac{3}{4}$  (D)  $\frac{6}{7}$

**Solution :**

$$\begin{array}{cccc} \frac{1}{2} \curvearrowright 1 \times 2 & \frac{5}{7} \curvearrowright 2 \times 1 & \frac{3}{4} \curvearrowright 1 \times 2 & \frac{6}{7} \curvearrowright 1 \times 2 \\ \frac{2}{4} \curvearrowright 2 & \frac{5}{7} \curvearrowright 2 & \frac{6}{8} \curvearrowright 2 & \frac{12}{14} \curvearrowright 2 \end{array}$$

(જ્યારે તફાવત સમાન હોય ત્યારે જેનો અંશ મોટો તે જવાબ બને.)

∴ (D)  $\frac{6}{7}$

(133) નીચેના અપૂર્ણાંકમાંથી સૌથી નાનો અપૂર્ણાંક કયો ?

$$\frac{5}{11}, \frac{7}{12}, \frac{8}{13}, \frac{9}{17}$$

(A)  $\frac{8}{13}$  (B)  $\frac{5}{11}$  (C)  $\frac{9}{17}$  (D)  $\frac{7}{12}$

**Solution :**

$$= 5, 6, 8$$

$$= 8 \times 3 \times 5$$

= 120 લ.સા.અ.

$$\frac{5}{11} \curvearrowright 6 \times 20 \quad \frac{7}{12} \curvearrowright 5 \times 24 \quad \frac{8}{13} \curvearrowright 5 \times 24 \quad \frac{9}{17} \curvearrowright 8 \times 15$$

$$\frac{100}{220} \quad \frac{168}{288} \quad \frac{192}{312} \quad \frac{135}{255}$$

⇒ જ્યારે તફાવત સમાન હોય ત્યારે જેનો અંશ નાનો તે નાનો અપૂર્ણાંક બને.

∴ (B)  $\frac{5}{11}$

(134) નીચે આપેલા અપૂર્ણાંકોમાંથી સૌથી મોટો અને સૌથી નાનો અપૂર્ણાંક વચ્ચેનો તફાવત શોધો.

$$\frac{5}{11}, \frac{5}{7}, \frac{3}{8}, \frac{6}{11}$$

(A)  $\frac{17}{56}$  (B)  $\frac{19}{56}$  (C)  $\frac{1}{7}$  (D)  $\frac{23}{56}$

**Solution :**

⇒ 6, 2, 5, 5 = 30 લ.સા.અ.

$$\begin{array}{cccc} \frac{5}{11} \curvearrowright 6 \times 5 & \frac{5}{7} \curvearrowright 2 \times 15 & \frac{3}{8} \curvearrowright 5 \times 6 & \frac{6}{11} \curvearrowright 5 \times 6 \\ 25 & \boxed{75} & \boxed{18} & 36 \end{array}$$

⇒  $\frac{5}{7} - \frac{3}{8} = \frac{40 - 21}{56} = \frac{19}{56}$

∴ (B)  $\frac{19}{56}$

(135)  $\frac{5}{9}, \frac{7}{11}, \frac{8}{15}$  અને  $\frac{11}{17}$  માંથી સૌથી મોટો અને સૌથી નાનો અપૂર્ણાંક વચ્ચેનો તફાવત શોધો.

(A)  $\frac{29}{255}$  (B)  $\frac{8}{99}$  (C)  $\frac{1}{45}$  (D)  $\frac{17}{165}$

**Solution :**

⇒ 4 4 7 6

⇒  $7 \times 6 \times 2 = 84$

$$\begin{array}{cccc} \frac{5}{9} \curvearrowright 4 \times 21 & \frac{7}{11} \curvearrowright 4 \times 21 & \frac{8}{15} \curvearrowright 7 \times 12 & \frac{11}{17} \curvearrowright 6 \times 14 \\ 105 & 147 & \boxed{96} & \boxed{154} \end{array}$$

⇒  $\frac{11}{17} - \frac{8}{15} = \frac{165 - 136}{255} = \frac{29}{255}$

∴ (A)  $\frac{29}{255}$

(136) નીચેનામાંથી નાની સંખ્યા કઈ ?

$$\sqrt{3}, \sqrt[3]{2}, \sqrt{2} \text{ અને } \sqrt[3]{4}$$

(A)  $\sqrt{2}$  (B)  $\sqrt[3]{2}$  (C)  $\sqrt{3}$  (D)  $\sqrt[3]{4}$

**Solution :**

$$= 2 \times 3 = 6$$

$$\begin{array}{cccc}
 (3) \frac{1}{2} \times \frac{3}{6} & (2) \frac{1}{8} \times \frac{2}{6} & (2) \frac{1}{2} \times \frac{3}{6} & (4) \frac{1}{8} \times \frac{2}{6} \\
 3^3 & 2^2 & 2^3 & 4^2 \\
 \downarrow & \downarrow & \downarrow & \downarrow \\
 27 & 4 & 8 & 16 \\
 & = \sqrt[3]{2} & & 
 \end{array}$$

$$\therefore (B) \sqrt[3]{2}$$

(137)  $2\frac{1}{2}$ ,  $3\frac{1}{3}$ ,  $4\frac{1}{4}$ ,  $6\frac{1}{6}$  અને  $12\frac{1}{12}$  માંથી મોટી સંખ્યા કઈ છે ?

(A)  $2\frac{1}{2}$  (B)  $3\frac{1}{3}$  (C)  $4\frac{1}{4}$  (D)  $6\frac{1}{6}$

**Solution :**

$\Rightarrow 2, 3, 4, 6, 12 = 12$  લ.સ.અ.

$$\begin{array}{ccccc}
 \frac{1}{2} \times \frac{6}{12} & \frac{1}{3} \times \frac{4}{12} & \frac{1}{4} \times \frac{3}{12} & \frac{1}{6} \times \frac{2}{12} & \frac{1}{12} \times \frac{1}{12} \\
 2^6 & 3^4 & 4^3 & 6^2 & 12^1 \\
 & \uparrow & & & \\
 64 & 81 & 64 & 36 & 12 \\
 & = 3^3 & & & 
 \end{array}$$

$$\therefore (B) 3\frac{1}{3}$$

(138) નીચેનામાંથી કઈ સંખ્યા પૂર્ણવર્ગ છે ?

(A) 5778 (B) 5777 (C) 5776 (D) 5772

**Solution :**

$\Rightarrow$  પૂર્ણવર્ગ સંખ્યાની એકમનો અંક  $\rightarrow 0, 1, 4, 5, 6, 9$

$\Rightarrow$  પૂર્ણવર્ગ સંખ્યા ન હોય તેમનો એકમનો અંક :

$\rightarrow 2, 3, 7, 8$

$\Rightarrow$  પૂર્ણવર્ગ સંખ્યાનો સરવાળો કરતાં એકમનો અંક :

$\rightarrow 1, 9, 4, 7$  આવે.

$$\therefore (C) 5776$$

(139)  $\frac{4}{5}, \frac{7}{8}, \frac{6}{7}, \frac{5}{6}$  ને ચડતા ક્રમમાં ગોઠવો.

(A)  $\frac{4}{5}, \frac{7}{8}, \frac{6}{7}, \frac{5}{6}$  (B)  $\frac{7}{8}, \frac{6}{7}, \frac{5}{6}, \frac{4}{5}$   
 (C)  $\frac{4}{5}, \frac{5}{6}, \frac{6}{7}, \frac{7}{8}$  (D)  $\frac{5}{6}, \frac{6}{7}, \frac{7}{8}, \frac{4}{5}$

**Solution :**

$\Rightarrow$  તમામનો લઘુગુણક 1 એટલે

$\Rightarrow$  સીધી જ સંખ્યાઓ અંશના આધારે ગોઠવો.

$$\therefore (C) \frac{4}{5}, \frac{5}{6}, \frac{6}{7}, \frac{7}{8}$$

(140)  $\frac{3}{5}, \frac{7}{9}, \frac{11}{13}$  ને ઉતરતા ક્રમમાં ગોઠવો.

(A)  $\frac{3}{5}, \frac{7}{9}, \frac{11}{13}$  (B)  $\frac{7}{9}, \frac{3}{5}, \frac{11}{13}$   
 (C)  $\frac{11}{13}, \frac{7}{9}, \frac{3}{5}$  (D)  $\frac{11}{13}, \frac{3}{5}, \frac{7}{9}$

**Solution :**

$\Rightarrow$  લઘુગુણક : 2

$$\therefore \frac{11}{13} > \frac{7}{9} > \frac{3}{5}$$

$$\therefore (C) \frac{11}{13}, \frac{7}{9}, \frac{3}{5}$$

### Type # 16

(141)  $\left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{4}\right) \dots \left(1 - \frac{1}{23}\right) = ?$

(A)  $\frac{1}{23}$  (B)  $\frac{1}{22}$  (C)  $\frac{1}{3}$  (D)  $\frac{1}{2}$

**Solution :**

$$\Rightarrow \left(\frac{1}{2}\right) \times \left(\frac{2}{3}\right) \times \left(\frac{3}{4}\right) \dots \left(\frac{22}{23}\right) = \left(\frac{1}{23}\right)$$

$$\therefore (A) \frac{1}{23}$$

(142)  $\left[\left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{5}\right) \left(1 - \frac{1}{6}\right) \dots \left(1 - \frac{1}{99}\right) \left(1 - \frac{1}{100}\right)\right]$

(A)  $\frac{2}{99}$  (B)  $\frac{1}{25}$  (C)  $\frac{1}{50}$  (D)  $\frac{1}{100}$

**Solution :**

$$\Rightarrow \left(\frac{2}{3}\right) \left(\frac{3}{4}\right) \left(\frac{4}{5}\right) \left(\frac{5}{6}\right) \dots \left(\frac{98}{99}\right) \left(\frac{99}{100}\right)$$

$$= \frac{2}{100}$$

$$= \frac{1}{50}$$

$$\therefore (C) \frac{1}{50}$$

(143)  $\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \frac{1}{42} + \frac{1}{56} + \frac{1}{72}$

(A)  $\frac{1}{2}$  (B) 0 (C)  $\frac{8}{9}$  (D)  $\frac{1}{2520}$

**Solution :**

$$\Rightarrow \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \frac{1}{5 \times 6} + \frac{1}{6 \times 7} + \frac{1}{7 \times 8} + \frac{1}{8 \times 9}$$

$$\therefore \text{સૂત્ર : } \frac{1}{\text{તફાવત}} [\text{પ્રથમ પદ} - \text{છેલ્લું પદ}]$$

રીત : 1

⇒ તમામ અંશનો સરવાળો

⇒ છેદમાં અંતિમ પદ અને પ્રથમ સંખ્યાનો ગુણાકાર

$$\therefore \frac{8}{1 \times 9} = \frac{8}{9}$$

રીત : 2

$$\therefore \frac{1}{1} \left[ \frac{1}{1} - \frac{1}{9} \right] = 1 - \frac{1}{9} = \frac{8}{9}$$

$$\therefore (C) \frac{8}{9}$$

$$(144) \frac{5}{2^2 \cdot 3^2} + \frac{7}{3^2 \cdot 4^2} + \frac{9}{4^2 \cdot 5^2} + \frac{11}{5^2 \cdot 6^2} + \frac{13}{6^2 \cdot 7^2} + \frac{15}{7^2 \cdot 8^2} + \frac{17}{8^2 \cdot 9^2} + \frac{19}{9^2 \cdot 10^2}$$

(A)  $\frac{1}{100}$  (B)  $\frac{6}{25}$  (C)  $\frac{101}{100}$  (D) 1

Solution :

$$\begin{aligned} &= \frac{5}{2^2 \cdot 3^2} + \frac{7}{3^2 \cdot 4^2} + \frac{9}{4^2 \cdot 5^2} + \frac{11}{5^2 \cdot 6^2} + \frac{13}{6^2 \cdot 7^2} + \frac{15}{7^2 \cdot 8^2} + \frac{17}{8^2 \cdot 9^2} + \frac{19}{9^2 \cdot 10^2} \\ &= \frac{5}{4 \times 9} + \frac{7}{9 \times 16} + \frac{9}{16 \times 25} + \frac{11}{25 \times 36} + \frac{13}{36 \times 49} + \frac{15}{49 \times 64} + \frac{17}{64 \times 81} + \frac{19}{81 \times 100} \\ &= \text{પ્રથમ પદ} - \text{છેલ્લું પદ} \\ &= \frac{1}{4} - \frac{1}{100} \\ &= \frac{100-4}{400} \\ &= \frac{96}{400} \\ &= \frac{6}{25} \end{aligned}$$

$$\therefore (B) \frac{6}{25}$$

$$(145) \frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \frac{1}{7 \times 10} + \frac{1}{10 \times 13} + \frac{1}{13 \times 16}$$

(A)  $\frac{1}{32}$  (B)  $\frac{7}{16}$   
(C)  $\frac{5}{16}$  (D)  $\frac{1}{210}$

Solution :

$$\begin{aligned} \Rightarrow & \frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \frac{1}{7 \times 10} + \frac{1}{10 \times 13} + \frac{1}{13 \times 16} \\ &= \frac{1}{3} \left[ \frac{1}{1} - \frac{1}{16} \right] \quad \text{તફાવત} \\ &= \frac{1}{3} \left[ \frac{16-1}{16} \right] \\ &= \frac{1}{3} \times \frac{15}{16} \\ &= \frac{5}{16} \end{aligned}$$

(અથવા)

$$= \frac{5}{16 \times 1} = \frac{5}{16}$$

$$\therefore (C) \frac{5}{16}$$

$$(146) \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \frac{1}{7 \cdot 9} + \frac{1}{9 \cdot 11} + \frac{1}{11 \cdot 13} + \frac{1}{13 \cdot 15}$$

(A)  $\frac{11}{18}$  (B)  $\frac{24}{53}$  (C)  $\frac{56}{53}$  (D)  $\frac{2}{15}$

Solution :

$$\begin{aligned} \Rightarrow & \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \frac{1}{7 \times 9} + \frac{1}{9 \times 11} + \frac{1}{11 \times 13} + \frac{1}{13 \times 15} \\ &= \frac{1}{2} \left[ \frac{1}{3} - \frac{1}{15} \right] \quad \text{તફાવત} \\ &= \frac{1}{2} \left[ \frac{15-3}{45} \right] \\ &= \frac{1}{2} \times \frac{12}{45} \\ &= \frac{2}{15} \quad \text{(અથવા)} \\ &= \frac{2}{15 \times 1} = \frac{2}{15} \\ &\therefore (D) \frac{2}{15} \end{aligned}$$

$$(147) \frac{1}{2 \cdot 5} + \frac{1}{5 \cdot 8} + \frac{1}{8 \cdot 11} + \frac{1}{11 \cdot 14} + \frac{1}{14 \cdot 17} + \frac{1}{17 \cdot 20} + \frac{1}{20 \cdot 23} + \frac{1}{23 \cdot 26} + \frac{1}{26 \cdot 29} = ?$$

(A)  $\frac{11}{18}$  (B)  $\frac{24}{53}$  (C)  $\frac{56}{53}$  (D)  $\frac{9}{58}$

Solution :

$$\begin{aligned} \Rightarrow & \frac{1}{2 \cdot 5} + \frac{1}{5 \cdot 8} + \frac{1}{8 \cdot 11} + \frac{1}{11 \cdot 14} + \frac{1}{14 \cdot 17} + \frac{1}{17 \cdot 20} + \frac{1}{20 \cdot 23} + \frac{1}{23 \cdot 26} + \frac{1}{26 \cdot 29} = ? \\ &= \frac{1}{3} \left[ \frac{1}{2} - \frac{1}{29} \right] \quad \text{તફાવત} \\ &= \frac{1}{3} \left[ \frac{29-2}{58} \right] = \frac{1}{3} \times \frac{27}{58} = \frac{9}{58} \\ &\quad \text{(અથવા)} \\ &= \frac{9}{29 \times 2} = \frac{9}{58} \\ &\therefore (D) \frac{9}{58} \end{aligned}$$

(148)  $\frac{1}{1 \cdot 5} + \frac{1}{5 \cdot 9} + \frac{1}{9 \cdot 13} + \frac{1}{13 \cdot 17} + \frac{1}{17 \cdot 21} = ?$

- (A)  $\frac{11}{18}$  (B)  $\frac{5}{21}$   
(C)  $\frac{56}{53}$  (D)  $\frac{14}{65}$

**Solution :**

$$= \frac{1}{\underbrace{1 \times 5}_4} + \frac{1}{\underbrace{5 \times 9}_4} + \frac{1}{9 \times 13} + \frac{1}{13 \times 17} + \frac{1}{17 \times 21}$$

તફાવત

$$= \frac{1}{4} \left[ \frac{1}{1} - \frac{1}{21} \right] = \frac{1}{4} \left[ \frac{21-1}{21} \right] = \frac{1}{4} \times \frac{20}{21} = \frac{5}{21}$$

(અથવા)

$$= \frac{5}{21 \times 1} = \frac{5}{21}$$

$$\therefore \text{(B) } \frac{5}{21}$$

(149)  $\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \frac{1}{42} + \frac{1}{56} + \frac{1}{72} + \frac{1}{90} = ?$

- (A)  $\frac{8}{9}$  (B)  $\frac{24}{53}$   
(C)  $\frac{56}{53}$  (D)  $\frac{9}{10}$

**Solution :**

$$\Rightarrow \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \frac{1}{42} + \frac{1}{56} + \frac{1}{72} + \frac{1}{90}$$

$$= 1 \times 2 \quad 2 \times 3 \quad 3 \times 4 \quad \dots \dots \dots 8 \times 9 \quad 9 \times 10$$

(1) (1) તફાવત

$$= \frac{1}{1} - \frac{1}{10} = \frac{10-1}{10} = \frac{9}{10}$$

(અથવા)

$$= \frac{9}{10 \times 1} = \frac{9}{10}$$

$$\therefore \text{(D) } \frac{9}{10}$$

(150)  $\frac{1}{30} + \frac{1}{42} + \frac{1}{56} + \frac{1}{72} + \frac{1}{90} + \frac{1}{110} + \frac{1}{132} = ?$

- (A)  $\frac{11}{18}$  (B)  $\frac{7}{60}$   
(C)  $\frac{56}{53}$  (D)  $\frac{14}{65}$

**Solution :**

$$\Rightarrow \frac{1}{30} + \frac{1}{42} + \frac{1}{56} + \frac{1}{72} + \frac{1}{90} + \frac{1}{110} + \frac{1}{132}$$

$$5 \times 6 \quad 6 \times 7 \quad 7 \times 8 \quad 8 \times 9 \quad 9 \times 10 \quad 10 \times 11 \quad 11 \times 12$$

$$= \frac{7}{12 \times 5} = \frac{7}{60}$$

$$\therefore \text{(B) } \frac{7}{60}$$

(151)  $\frac{1}{30} + \frac{1}{42} + \frac{1}{56} + \frac{1}{72} + \frac{1}{90} + \frac{1}{110} + \frac{1}{132} + \frac{1}{156} + \frac{1}{182} = ?$

- (A)  $\frac{9}{70}$  (B)  $\frac{24}{53}$   
(C)  $\frac{56}{53}$  (D)  $\frac{14}{65}$

**Solution :**

$$\frac{1}{30} + \frac{1}{42} + \frac{1}{56} + \frac{1}{72} + \frac{1}{90} + \frac{1}{110} + \frac{1}{132} + \frac{1}{156} + \frac{1}{182}$$

$$5 \times 6 \quad 6 \times 7 \quad 7 \times 8 \quad \dots \dots \dots 12 \times 13 \quad 13 \times 14$$

$$= \frac{9}{14 \times 5} = \frac{9}{70}$$

$$\therefore \text{(A) } \frac{9}{70}$$

(152)  $\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \frac{1}{3 \cdot 4 \cdot 5} + \frac{1}{4 \cdot 5 \cdot 6} = ?$

- (A)  $\frac{11}{18}$  (B)  $\frac{24}{53}$   
(C)  $\frac{56}{53}$  (D)  $\frac{7}{30}$

**Solution :**

$$= \frac{1}{4} \left( \frac{4 \times 7}{5 \times 6} \right)$$

- $1/4$  ફરજિયાત તફાવત લખવો

- અંતિમ પદની પ્રથમ સંખ્યા (4) અંશમાં લખવી.

- ત્યારબાદની બંને સંખ્યાને છેદમાં ગુણાકારમાં લખવી.

- છેદમાં રહેલી અંતિમ સંખ્યા (6) ને 1 ઉમેરી (7) અંશમાં આપેલી સંખ્યા સાથે ગુણવી.

$$= \frac{1}{4} \left( \frac{4 \times 7}{5 \times 6} \right)$$

$$= \frac{7}{30}$$

$$\therefore \text{(D) } \frac{7}{30}$$

(153)  $\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \frac{1}{3 \cdot 4 \cdot 5} + \frac{1}{4 \cdot 5 \cdot 6} + \frac{1}{5 \cdot 6 \cdot 7} = ?$

- (A)  $\frac{11}{18}$  (B)  $\frac{24}{53}$   
(C)  $\frac{5}{21}$  (D)  $\frac{7}{30}$

**Solution :**

$$\Rightarrow \frac{1}{1 \times 2 \times 3} + \frac{1}{2 \times 3 \times 4} + \frac{1}{3 \times 4 \times 5} + \frac{1}{4 \times 5 \times 6} + \frac{1}{5 \times 6 \times 7}$$

$$= \frac{1}{4} \left( \frac{5 \times \cancel{2}}{6 \times 7} \right) = \frac{5 \times \cancel{2}}{\cancel{42}} = \frac{5}{21}$$

$$\therefore \text{(C) } \frac{5}{21}$$



## Type # 17

(154)  $0.5 \div 0.125 = ?$

- (A) 0.625 (B) 0.575  
(C) 0.0575 (D) 4

Solution :

$$\Rightarrow \frac{0.5}{0.125} = \frac{\frac{5}{10}}{\frac{125}{1000}} = \frac{5 \times 100}{125 \times 10} = \frac{500}{1250} = \frac{4}{10} = 0.4$$

 $\therefore$  (D) 4

(155)  $23 \times (64 - 24) \div 100 = ?$

- (A) 92 (B) 8.2  
(C) 9.2 (D) 82

Solution :

$$\Rightarrow 23 \times \frac{40}{100} = \frac{92}{10} = 9.2$$

 $\therefore$  (C) 9.2

(156)  $\frac{4}{8} \div \left(\frac{1}{2} - \frac{1}{3}\right) \times \frac{4}{6} = ?$

- (A) 1 (B)  $1\frac{1}{2}$   
(C)  $2\frac{1}{2}$  (D) 2

Solution :

$$\begin{aligned} \Rightarrow \frac{4}{8} \div \left(\frac{3-2}{6}\right) \times \frac{4}{6} &= \frac{4}{8} \div \frac{1}{6} \times \frac{4}{6} \\ &= \frac{4}{8} \times \frac{6}{1} \times \frac{4}{6} \\ &= \frac{4}{2} \times \frac{4}{2} \\ &= 2 \end{aligned}$$

 $\therefore$  (D) 2

(157)  $11 \times 5 + 8 - 3 = 2 \times 2 \times 3 \times ?$

- (A) 62 (B) 12 (C) 60 (D) 5

Solution :

$$\begin{aligned} \Rightarrow 11 \times 5 + 8 - 3 &= 2 \times 2 \times 3 \times ? \\ 55 + 8 - 3 &= 12x \\ 63 - 3 &= 12x \\ x &= \frac{60}{12} \\ x &= 5 \end{aligned}$$

 $\therefore$  (D) 5

(158)  $\frac{(589+187)^2 - (589-187)^2}{589 \times 187} = ?$

- (A) 4 (B) 766 (C) 402 (D)  $\frac{201}{388}$

Solution :

$$\begin{aligned} \Rightarrow \frac{(589+187)^2 - (589-187)^2}{589 \times 187} \\ &= \frac{(a+b)^2 - (a-b)^2}{a \times b} \quad (a = 589, b = 187) \\ &= \frac{a^2 + 2ab + b^2 - a^2 + 2ab - b^2}{ab} \\ &= \frac{4ab}{ab} \\ &= 4 \end{aligned}$$

 $\therefore$  (A) 4

(159)  $90 \div 6 \times (24 - 8 \div 4) \div 3 = ?$

- (A) 18 (B) 20 (C) 110 (D) 16

Solution :

$$\begin{aligned} \Rightarrow 90 \div 6 \times (24 - 2) \div 3 \\ &= 90 \div 6 \times 22 \div 3 \\ &= \frac{90}{6} \times \frac{22}{3} \\ &= 15 \times \frac{22}{3} \\ &= 110 \end{aligned}$$

 $\therefore$  (C) 110

(160)  $15.5 \times 1.2 = ?$

- (A) 18.06 (B) 18.6 (C) 1.86 (D) 186

Solution :

$$\Rightarrow \frac{155}{10} \times \frac{12}{10} = \frac{1860}{100} = 18.6$$

 $\therefore$  (B) 18.6

(161)  $\frac{11}{8} \times \frac{56}{33} = ?$

- (A)  $3\frac{1}{7}$  (B)  $7\frac{1}{3}$  (C)  $3\frac{1}{2}$  (D)  $2\frac{1}{3}$

Solution :

$$\Rightarrow \frac{11}{8} \times \frac{56}{33} = \frac{11}{8} \times \frac{7}{3} = \frac{7}{3} = 2\frac{1}{3}$$

 $\therefore$  (D)  $2\frac{1}{3}$

(162)  $(24 \times 36 \times 24) \div (2\sqrt{1296} \div 2)$

- (A) 526 (B) 276 (C) 512 (D) 576

Solution :

$$\begin{aligned}
 & (24 \times 36 \times 24) \div \left( \frac{2\sqrt{1296}}{2} \right) \\
 & = (24 \times \cancel{36} \times 24) \div \cancel{36} \\
 & = 24 \times 24 \\
 & = 576
 \end{aligned}$$

$$\begin{array}{r}
 36 \\
 3 \overline{)1296} \\
 \underline{+3} \quad 9 \\
 66 \quad 0396 \\
 \underline{\phantom{00}396} \\
 000
 \end{array}$$

$\therefore (D) 576$

(163)  $1.8 \div 1200 = ?$

- (A) 0.15 (B) 0.015
- 
- (C) 0.00015 (D) 0.0015

Solution :

$$\begin{aligned}
 & \frac{18}{1200} \\
 & = \frac{18}{12 \times 100} \\
 & = \frac{3}{200} \\
 & = \frac{3}{2} \times \frac{1}{100} \\
 & = \frac{1.5}{100} \\
 & = 0.0015
 \end{aligned}$$

$\therefore (D) 0.0015$

(164)  $9.87 + 6.54 + 3.21 = ?$

- (A) 19.62 (B) 18.62 (C) 19.72 (D) 18.52

Solution :

$\Rightarrow 9.87 + 6.54 + 3.21 = 19.62$

$\therefore (A) 19.62$

(165) 525 ની 28% કેટલી ?

- (A) 154 (B) 147 (C) 140 (D) 133

Solution :

$$\Rightarrow \frac{525 \times \cancel{28}}{\cancel{100}} = 21 \times 7 = 147$$

$\therefore (B) 147$

(166)  $10.747 + 6.59 + 2.26 = ?$

- (A) 19.597 (B) 18.597
- 
- (C) 19.697 (D) 18.697

Solution :

$\therefore 10.747 + 6.59 + 2.26 = 19.597$

$\therefore (A) 19.597$

(167) જો  $2\frac{1}{2} + 3\frac{1}{4} + x + 2\frac{1}{3} = 13\frac{5}{12}$  તો  $x$  ની કિંમત શોધો.

- (A)
- $4\frac{3}{8}$
- (B)
- $11\frac{9}{11}$
- 
- (C)
- $4\frac{2}{3}$
- (D)
- $5\frac{1}{3}$

Solution :

$\Rightarrow \frac{5}{2} + \frac{13}{4} + x + \frac{7}{3} = \frac{161}{12}$

$\therefore \text{લ.સ.અ.} = 12$

$\therefore \frac{30 + 39 + 12x + 28}{12} = \frac{161}{12}$

$\therefore \frac{97 + 12x}{12} = \frac{161}{12}$

$\therefore 97 + 12x = 161$

$\therefore 12x = 161 - 97$

$\therefore 12x = 64$

$\therefore x = \frac{64}{12} = \frac{16}{3}$

$\therefore x = 5\frac{1}{3}$

$\therefore (D) 5\frac{1}{3}$

(168)  $1\frac{1}{2} + 11\frac{1}{2} + 111\frac{1}{2} + 1111\frac{1}{2}$

- (A) 1236 (B) 3456
- 
- (C) 618 (D) 617

Solution :

$= (1 + 11 + 111 + 1111) + \left( \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} \right)$

$[\therefore \text{છેદ સરખા તો અંશનો સરવાળો}]$

$= 1234 + \frac{2}{2}$

$= 1236$

$\therefore (A) 1236$

## Type # 18

□ સમજૂતી :

⇒ આધાર સરખા હોય અને ગુણાકાર કરવાનો હોય તો ઘાતનો સરવાળો થાય.

$$a^m \times a^n = a^{m+n}$$

⇒ આધાર સરખા હોય અને ભાગકાર કરવાનો હોય તો ઘાતની બાદબાકી થાય.

$$a^m \div a^n = a^{m-n}$$

⇒ ઘાતની ઘાત કૌંસમાં હોય તો બંને ઘાતનો ગુણાકાર થાય.

$$(a^m)^n = a^{mn}$$

(169)  $x$  ની કિંમત શોધો :  $8^{2.3} \times 4^{0.8} \times 16^{0.4} = 2^x$

(A) 11.4

(B) 10.2

(C) 7.7

(D) 10.1

Solution :

$$\therefore (2^3)^{2.3} \times (2^2)^{0.8} \times (2^4)^{0.4} = 2^x$$

$$\therefore 2^{6.9} \times 2^{1.6} \times 2^{1.6} = 2^x$$

$$\therefore 2^{6.9+1.6+1.6} = 2^x$$

$$\therefore 2^{10.1} = 2^x$$

$$\therefore x = 10.1$$

(આધાર સરખા હોવાથી બંનેની ઘાતની કિંમત પણ સમાન આવે.)

$$\therefore (D) 10.1$$

(170)  $95^{2.4} \times 95^{1.3} \div 95^{0.9989} = 95^m$   
તો  $m$  ની કિંમત શોધો.

(A) 2.7011

(B) 0.027011

(C) 0.27011

(D) 27.011

Solution :

$$= 95^{2.4} \times 95^{1.3} \div 95^{0.9989} = 95^m$$

$$= 2.4 + 1.3$$

$$3.7000$$

$$= 0.9989$$

$$2.7011$$

$$\therefore 95^{2.7011} = 95^m$$

$$\therefore m = 2.7011$$

$$\therefore (A) 2.7011$$

$$(171) \frac{(1024^{0.13} \times 1024^{0.07})}{(7^{0.25} \times 49^{0.075} \times 343^{0.2})}$$

$$(A) \frac{3}{49}$$

$$(B) \frac{3}{16}$$

$$(C) \frac{3}{52}$$

$$(D) \frac{4}{7}$$

Solution :

$$= \frac{(2^{10})^{0.13} \times (2^{10})^{0.07}}{7^{0.25} \times (7^2)^{0.075} \times (7^3)^{0.2}}$$

$$= \frac{2^{1.3} \times 2^{0.7}}{7^{0.25} \times 7^{0.15} \times 7^{0.60}}$$

$$= \frac{2^2}{7^1}$$

$$= \frac{4}{7}$$

$$\therefore (D) \frac{4}{7}$$

$$(172) \sqrt[8]{\sqrt[3]{(81^2)^3}}$$

(A) 5

(B) 9

(C) 11

(D) 3

Solution :

$$\Rightarrow \sqrt[8]{\sqrt[3]{(81^2)^3}} = 81^{\cancel{2} \times \cancel{3} \times \frac{1}{\cancel{3}} \times \frac{1}{\cancel{8}}}$$

$$= 81^{\frac{1}{4}}$$

$$= (3^4)^{1/4}$$

$$= 3^1$$

$$= 3$$

$$\therefore (D) 3$$

## Type # 19

□ સમજૂતી :

$$\Rightarrow \therefore (a+b)^2 = a^2 + 2ab + b^2$$

$$\therefore \left(x + \frac{1}{x}\right)^2 = x^2 + 2 \times \cancel{x} \times \frac{1}{\cancel{x}} + \frac{1}{x^2}$$

$$= x^2 + \frac{1}{x^2} + 2$$

$$\Rightarrow \therefore (a-b)^2 = a^2 - 2ab + b^2$$

$$\Rightarrow \left(x - \frac{1}{x}\right)^2 = x^2 - 2 \times \cancel{x} \times \frac{1}{\cancel{x}} + \frac{1}{x^2}$$

$$= x^2 + \frac{1}{x^2} - 2$$

$$\Rightarrow \therefore a^2 - b^2 = (a+b)(a-b)$$

$$= x^2 - \frac{1}{x^2} = \left(x + \frac{1}{x}\right)\left(x - \frac{1}{x}\right)$$

$$\Rightarrow \therefore a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$\therefore x^3 + \frac{1}{x^3}$$

$$= \left(x + \frac{1}{x}\right) \left(x^2 - \cancel{x} \times \frac{1}{\cancel{x}} + \frac{1}{x^2}\right)$$

$$= \left(x + \frac{1}{x}\right) \left(x^2 + \frac{1}{x^2} - 1\right)$$

$$\Rightarrow \therefore a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$\therefore x^3 - \frac{1}{x^3}$$

$$= \left(x - \frac{1}{x}\right) \left(x^2 + \cancel{x} \times \frac{1}{\cancel{x}} + \frac{1}{x^2}\right)$$

$$= \left(x - \frac{1}{x}\right) \left(x^2 + \frac{1}{x^2} + 1\right)$$

(173) જો  $x + \frac{1}{x} = 4$  હોય, તો  $x^2 + \frac{1}{x^2} = ?$

(A) 62

(B) 14

(C) 16

(D) 36

**Solution :**

**ચિત્ર : 1**

$$\left(x + \frac{1}{x}\right)^2 = (4)^2 \quad \{\text{વર્ગ કરીને } -2 \text{ કરવા}\}$$

$$\therefore x^2 + 2 \times \cancel{x} \times \frac{1}{\cancel{x}} + \frac{1}{x^2} = 16$$

$$\therefore x^2 + \frac{1}{x^2} + 2 = 16$$

$$\therefore x^2 + \frac{1}{x^2} = 16 - 2$$

$$\therefore x^2 + \frac{1}{x^2} = 14$$

**ચિત્ર : 2**

$$\Rightarrow K^2 - 2 = (4)^2 - 2 = 16 - 2 = 14$$

$$\therefore (B) 14$$

(174) જો  $x + \frac{1}{x} = \sqrt{5}$  હોય, તો  $x^3 + \frac{1}{x^3} = ?$

(A)  $3\sqrt{5}$

(B)  $4\sqrt{5}$

(C)  $2\sqrt{5}$

(D)  $5\sqrt{5}$

**Solution :**

**ચિત્ર : 1**

$$= \left(x + \frac{1}{x}\right) \left(x^2 - \cancel{x} \times \frac{1}{\cancel{x}} + \frac{1}{x^2}\right)$$

$$= \left(x + \frac{1}{x}\right) \left(x^2 + \frac{1}{x^2} - 1\right)$$

$$= \sqrt{5}(3-1)$$

$$= \sqrt{5} \times 2$$

$$= 2\sqrt{5}$$

**ચિત્ર : 2**

$$\Rightarrow K^2 - 3K = (\sqrt{5})^3 - 3(\sqrt{5})$$

$$= \sqrt{5}(5-3)$$

$$= 2\sqrt{5}$$

$$\therefore (C) 2\sqrt{5}$$

(175) જો  $x + \frac{1}{x} = 3$  હોય, તો  $x^3 + \frac{1}{x^3} = ?$

(A) 27

(B) 36

(C) 24

(D) 18

**Solution :**

$$\Rightarrow x^2 + \frac{1}{x^2} \text{ માટે } \downarrow \text{ વર્ગ}$$

$$\downarrow - 2$$

$$(3)^2 = 9 - 2 = 7$$

$$\Rightarrow \left(x + \frac{1}{x}\right) \left(x^2 + \frac{1}{x^2} - 1\right) = (3)(7-1)$$

$$= 3 \times 6$$

$$= 18$$

(અથવા)

$$\therefore K^3 - 3K = (3)^3 - 3(3) = 27 - 9 = 18$$

$$\therefore (D) 18$$

(176) જો  $\sqrt{x} + \frac{1}{\sqrt{x}} = \sqrt{6}$  હોય, તો  $x^2 + \frac{1}{x^2} = ?$

(A) 60

(B) 14

(C) 16

(D) 36

**Solution :**

$$\Rightarrow \text{ચિત્ર : 1}$$

$$\therefore \sqrt{x} + \frac{1}{\sqrt{x}} = \sqrt{6}$$

$$\therefore x + \frac{1}{x} = 4$$

$$\therefore (4)^2 - 2 = 16 - 2 = 14 = x^2 + \frac{1}{x^2}$$

(અથવા)

⇒ **ધાર : 2**

$$\therefore K^2 - 2 = (\sqrt{6})^2 - 2 = 6 - 2 = 4 \rightarrow x + \frac{1}{x}$$

$$\therefore K^2 - 2 = (4)^2 - 2 = 16 - 2 = 14 \rightarrow x^2 + \frac{1}{x^2}$$

**∴ (B) 14**

(177) જો  $\sqrt{x} + \frac{1}{\sqrt{x}} = 2\sqrt{2}$  હોય, તો  $x^2 + \frac{1}{x^2} = ?$

(A) 34 (B) 64 (C) 36 (D) 32

**Solution :**

$$= 2\sqrt{2} \times 2\sqrt{2} = 4 \times 2 = 8$$

$$x + \frac{1}{x} = K^2 - 2 = (2\sqrt{2})^2 - 2 = 8 - 2 = 6$$

$$x^2 + \frac{1}{x^2} = K^2 - 2 = (6)^2 - 2 = 36 - 2 = 34$$

**∴ (A) 34**

(178) જો  $x + \frac{1}{x} = 8$  હોય, તો  $\frac{5x}{x^2 + 1 - 6x} = ?$

(A) 2.5 (B) 6 (C) 5 (D) 6.5

**Solution :**

$$= \frac{5\cancel{x} / \cancel{x}}{\frac{\cancel{x}^2}{\cancel{x}} + \frac{1}{x} - \frac{6\cancel{x}}{\cancel{x}}}$$

$$= \frac{5}{x + \frac{1}{x} - 6}$$

$$= \frac{5}{8 - 6}$$

$$= \frac{5}{2}$$

$$= 2.5$$

**∴ (A) 2.5**

(179) જો  $\sqrt{x} - \frac{1}{\sqrt{x}} = \sqrt{6}$  હોય, તો  $x^2 + \frac{1}{x^2} = ?$

(A) 62 (B) 40 (C) 54 (D) 66

**Solution :**

$$\Rightarrow \left( \sqrt{x} - \frac{1}{\sqrt{x}} \right) = \sqrt{6} = \left( \sqrt{x} - \frac{1}{\sqrt{x}} \right)^2 = (\sqrt{6})^2$$

$$\therefore x - 2\left(\sqrt{x}\right)\left(\frac{1}{\sqrt{x}}\right) + \frac{1}{x} = 6$$

$$\therefore 2x + \frac{1}{x} = 8$$

$$\therefore \left( x + \frac{1}{x} \right)^2 = (8)^2$$

$$\therefore x^2 + 2\left(x\right)\left(\frac{1}{x}\right) + \frac{1}{x^2} = 64$$

$$\therefore x^2 + 2 + \frac{1}{x^2} = 64$$

$$\therefore x^2 + \frac{1}{x^2} = 64 - 2$$

$$\therefore x^2 + \frac{1}{x^2} = 62$$

**∴ (A) 62**

(180) જો  $x - \frac{1}{x} = 13$  હોય, તો  $x^2 + \frac{1}{x^2} = ?$

(A) 171 (B) 169

(C) 167 (D) 165

**Solution :**

$$\begin{aligned} \Rightarrow K^2 + 2 &= (13)^2 + 2 \\ &= 169 + 2 \\ &= 171 \end{aligned}$$

(અથવા)

$$\therefore x - \frac{1}{x} = 13$$

$$\therefore \left( x - \frac{1}{x} \right)^2 = (13)^2$$

$$\therefore x^2 - 2 \times \cancel{x} \times \frac{1}{\cancel{x}} + \frac{1}{x^2} = 169$$

$$\therefore x^2 - 2 + \frac{1}{x^2} = 169$$

$$\therefore x^2 + \frac{1}{x^2} = 169 + 2$$

$$\therefore x^2 + \frac{1}{x^2} = 171$$

**∴ (A) 171**

(181) જો  $x - \frac{1}{x} = 4$  હોય, તો  $x^2 + \frac{1}{x^2} = ?$

(A) 14 (B) 16

(C) 18 (D) 20

**Solution :**

$$\begin{aligned} \Rightarrow K^2 + 2 &= (4)^2 + 2 \\ &= 16 + 2 \\ &= 18 \end{aligned}$$

**∴ (C) 18**



(182) જો  $x + \frac{1}{x} = 3$  હોય, તો  $x - \frac{1}{x} = ?$

- (A)  $\sqrt{2}$  (B)  $\sqrt{3}$   
(C)  $\sqrt{5}$  (D)  $\sqrt{6}$

**Solution :**

$$\therefore \sqrt{x^2 - 4} = \sqrt{9 - 4} = \sqrt{5}$$

$$\therefore \text{(C) } \sqrt{5}$$

(183) જો  $x + \frac{1}{x} = 5$  હોય, તો  $x - \frac{1}{x} = ?$

- (A)  $\sqrt{20}$  (B)  $\sqrt{21}$   
(C)  $\sqrt{25}$  (D)  $\sqrt{26}$

**Solution :**

$$\therefore \sqrt{x^2 - 4} = \sqrt{25 - 4} = \sqrt{21}$$

$$\therefore \text{(B) } \sqrt{21}$$

(184) જો  $x - \frac{1}{x} = 5$  હોય, તો  $x + \frac{1}{x} = ?$

- (A)  $\sqrt{24}$  (B)  $\sqrt{26}$   
(C)  $\sqrt{27}$  (D)  $\sqrt{29}$

**Solution :**

$$\therefore \sqrt{x^2 + 4} = \sqrt{25 + 4} = \sqrt{29}$$

$$\therefore \text{(D) } \sqrt{29}$$

(185) જો  $x - \frac{1}{x} = 4$  હોય, તો  $x + \frac{1}{x} = ?$

- (A)  $\sqrt{18}$  (B)  $\sqrt{19}$   
(C)  $\sqrt{20}$  (D)  $\sqrt{21}$

**Solution :**

$$\therefore \sqrt{x^2 + 4} = \sqrt{16 + 4} = \sqrt{20}$$

$$\therefore \text{(C) } \sqrt{20}$$

(186) જો  $a + \frac{1}{a} = 4$  હોય, તો  $a - \frac{1}{a} = ?$

- (A) 12 (B)  $2\sqrt{3}$   
(C)  $\sqrt{14}$  (D) 14

**Solution :**

$$\begin{aligned} \therefore \sqrt{a^2 - 4} &= \sqrt{12} \\ &= \sqrt{4 \times 3} \\ &= 2\sqrt{3} \end{aligned}$$

$$\therefore \text{(B) } 2\sqrt{3}$$

(187) જો  $x + \frac{1}{x} = 3$  હોય, તો  $x^2 - \frac{1}{x^2} = ?$

- (A)  $2\sqrt{3}$  (B)  $5\sqrt{3}$   
(C)  $3\sqrt{5}$  (D)  $4\sqrt{3}$

**Solution :**

$$\therefore x - \frac{1}{x} = \sqrt{9 - 4} = \sqrt{5}$$

$$\begin{aligned} \therefore x^2 - \frac{1}{x^2} &= \left(x + \frac{1}{x}\right)\left(x - \frac{1}{x}\right) \\ &= (3)(\sqrt{5}) \\ &= 3\sqrt{5} \end{aligned}$$

$$\therefore \text{(C) } 3\sqrt{5}$$

(188) જો  $x + \frac{1}{x} = 4$  હોય, તો  $x^3 + \frac{1}{x^3} = ?$

- (A) 52 (B) 66  
(C) 76 (D) 80

**Solution :**

$$\therefore ++ \rightarrow - \quad \text{થાપી}$$

$$\begin{aligned} \Rightarrow K^3 - 3K &= (4)^3 - 3(4) \\ &= 64 - 12 \\ &= 52 \end{aligned}$$

(અથવા)

$$\Rightarrow x^2 + \frac{1}{x^2} = 16 - 2 = 14$$

$$\begin{aligned} \Rightarrow x^3 + \frac{1}{x^3} &= \left(x + \frac{1}{x}\right)\left(x^2 - \cancel{x} \times \frac{1}{\cancel{x}} + \frac{1}{x^2}\right) \\ &= (4)(14 - 1) \\ &= 13 \times 4 \\ &= 52 \end{aligned}$$

$$\therefore \text{(A) } 52$$

(189) જો  $x - \frac{1}{x} = 5$  હોય, તો  $x^3 - \frac{1}{x^3} = ?$

- (A) 120 (B) 130 (C) 135 (D) 140

**Solution :**

$$\therefore -- \rightarrow + \quad \text{થાપી}$$

$$\Rightarrow x - \frac{1}{x} = 5$$

$$\Rightarrow (5)^2 + 2 = 25 + 2 = 27 = x^2 + \frac{1}{x^2}$$

$$\Rightarrow x^3 - \frac{1}{x^3} = \left(x - \frac{1}{x}\right)\left(x^2 + \cancel{x} \times \frac{1}{\cancel{x}} + \frac{1}{x^2}\right)$$

$$= (5) \left( x^2 + \frac{1}{x^2} + 1 \right)$$

$$= (5)(27+1)$$

$$= 5 \times 28$$

$$= 140$$

(અથવા)

$$\therefore K^3 + 3K = (5)^3 + 3(5)$$

$$= 125 + 15$$

$$= 140$$

$$\therefore (D) 140$$

(190) જો  $x + \frac{1}{x} = 3$  હોય, તો  $x^3 - \frac{1}{x^3} = ?$

(A)  $6\sqrt{5}$

(B)  $7\sqrt{5}$

(C)  $8\sqrt{5}$

(D)  $9\sqrt{5}$

Solution :

$$\Rightarrow \therefore x + \frac{1}{x} = 3$$

$$\Rightarrow \therefore x^2 + \frac{1}{x^2} = 9 - 2 = 7$$

$$\Rightarrow x - \frac{1}{x} = \sqrt{9-4} = \sqrt{5}$$

$$\Rightarrow x^3 - \frac{1}{x^3} = \left( x - \frac{1}{x} \right) \left( x^2 + \cancel{x} \times \frac{1}{\cancel{x}} + \frac{1}{x^2} \right)$$

$$= (\sqrt{5}) \left( x^2 + \frac{1}{x^2} + 1 \right)$$

$$= (\sqrt{5})(7+1)$$

$$= 8\sqrt{5}$$

$$\therefore (C) 8\sqrt{5}$$

(191) જો  $x + \frac{1}{x} = 3$  હોય, તો  $x^4 + \frac{1}{x^4} = ?$

(A) 44

(B) 46

(C) 40

(D) 47

Solution :

$$\Rightarrow x - \frac{1}{x} = 3$$

$$\Rightarrow x^2 + \frac{1}{x^2} \rightarrow 9 - 2 = 7$$

$$\Rightarrow x^4 + \frac{1}{x^4} \rightarrow 7^2 - 2 = 49 - 2 = 47$$

$$\therefore (D) 47$$

(192) જો  $x + \frac{1}{x} = 3$  હોય, તો  $x^5 + \frac{1}{x^5} = ?$

(A) 121

(B) 122

(C) 123

(D) 124

Solution :

$$\Rightarrow x^2 + \frac{1}{x^2} = 3^2 - 2 = 7$$

$$\Rightarrow x^3 + \frac{1}{x^3} = K^3 - 3K = 27 - 9 = 18$$

$$\Rightarrow x^5 + \frac{1}{x^5} = \left( x^2 + \frac{1}{x^2} \right) \left( x^3 + \frac{1}{x^3} \right) = 7 \times 18$$

$$\Rightarrow x^5 + x^2 \times \frac{1}{x^3} + \frac{1}{x^2} \times x^3 \times \frac{1}{x^5} = 126$$

$$x^5 + \left( \frac{1}{x} + x \right) + \frac{1}{x^5} = 126$$

$$126 - 3 = x^5 + \frac{1}{x^5}$$

$$123 = x^5 + \frac{1}{x^5}$$

$$\therefore (C) 123$$

(193) જો  $x + \frac{1}{x} = 3$  હોય, તો  $x^6 + \frac{1}{x^6} = ?$

(A) 321

(B) 322

(C) 323

(D) 324

Solution :

$$\Rightarrow x^3 + \frac{1}{x^3} = K^3 - 3K$$

$$= 3^3 - 3(3)$$

$$= 27 - 9$$

$$= 18$$

$$\Rightarrow 324 = \left( x^3 + \frac{1}{x^3} \right) \left( x^3 + \frac{1}{x^3} \right)$$

$$324 = x^6 + \cancel{x^3} + \frac{1}{\cancel{x^3}} + \frac{1}{\cancel{x^3}} \times \cancel{x^3} + \frac{1}{x^6}$$

$$324 = x^6 + 1 + 1 + \frac{1}{x^6}$$

$$324 - 2 = 322$$

$$\therefore (B) 322$$

## Type # 20

□ સમજૂતી :

+ મોટી સંખ્યા

- નાની સંખ્યા

$$(194) \sqrt{63 + 2\sqrt{63 + 2\sqrt{63 + 2\sqrt{63 \dots \infty}}}}$$

(A) 3 (B) 6 (C) 9 (D) 12

Solution :

⇒ તફાવત 2 હોવો જોઈએ.

$$\Rightarrow 63 = 7 \times 9$$

$$\therefore (C) 9$$

$$(195) \sqrt{63 - 2\sqrt{63 - 2\sqrt{63 - 2\sqrt{63 \dots \infty}}}}$$

(A) 3 (B) 6 (C) 9 (D) 7

Solution :

⇒ તફાવત 2 હોવો જોઈએ.

$$\Rightarrow 63 = 7 \times 9$$

$$\therefore (D) 7$$

$$(196) \sqrt{88 - 3\sqrt{88 - 3\sqrt{88 - 3\sqrt{88 \dots \infty}}}}$$

(A) 8 (B) 13 (C) 15 (D) 18

Solution :

⇒ તફાવત 3 હોવી જોઈએ.

$$\Rightarrow 88 = 8 \times 11$$

$$\therefore (A) 8$$

$$(197) \frac{\sqrt{72 - \sqrt{72 - \sqrt{72 \dots \infty}}}}{\sqrt{56 + \sqrt{56 + \sqrt{56 \dots \infty}}}}$$

(A) 7 (B) 8 (C) 9 (D) 1

Solution :

$$\Rightarrow 72 = 8 \times 9$$

$$\Rightarrow 56 = 7 \times 8$$

$$\therefore \frac{8}{8} = 1$$

$$\therefore (D) 1$$

$$(198) \frac{47}{2^5 \times 5^3} \text{ માં દશાંશ પછી કેટલા અંક આવે ?}$$

(A) 2 (B) 3 (C) 4 (D) 5

Solution :

⇒  $2^m$  અને  $5^n$  માં m અને n જેટલા અંક આવે. $2^m \times 5^n$  હોય ત્યારે જે ઘાત મોટી તેટલા અંક આવે.

$$\Rightarrow \frac{47}{2^5 \times 5^3} = 5$$

⇒ મોટી ઘાત = 5

⇒ દશાંશ પછી 5 અંક આવે.

$$\therefore (D) 5$$

$$(199) \frac{43}{2^3 \times 5^7} \text{ માં દશાંશ પછી કેટલા અંક આવે ?}$$

(A) 3 (B) 4 (C) 5 (D) 7

Solution :

⇒ મોટી ઘાત 7 છે.

$$\therefore (D) 7$$

$$(200) \frac{6}{120} \text{ માં દશાંશ પછી કેટલા અંક આવે ?}$$

(A) 1 (B) 2 (C) 3 (D) 4

Solution :

$$\Rightarrow \frac{6}{120} = \frac{1}{20} = \frac{1}{4 \times 5} = \frac{1}{2^2 \times 5^1} = 2$$

⇒ મોટી ઘાત = 2

$$\therefore (B) 2$$

$$(201) 7 + 2\sqrt{10} \text{ નું વર્ગમૂળ શોધો.}$$

$$(A) \sqrt{3} + \sqrt{4}$$

$$(B) \sqrt{6} - 1$$

$$(C) \sqrt{2} + \sqrt{5}$$

$$(D) 2 + \sqrt{5}$$

Solution :

$$\Rightarrow \sqrt{7 + 2\sqrt{10}}$$

$$= \sqrt{2 + 5 + 2\sqrt{2} \times \sqrt{5}}$$

$$= \sqrt{(\sqrt{2})^2 + (\sqrt{5})^2 + 2\sqrt{2} \times \sqrt{5}}$$

$$\Rightarrow (a + b)^2 = a^2 + 2ab + b^2$$

$$= \sqrt{(\sqrt{2} + \sqrt{5})^2}$$

$$= (\sqrt{2} + \sqrt{5})^{2 \times \frac{1}{2}}$$

$$= \sqrt{2} + \sqrt{5}$$

$$\therefore (C) \sqrt{2} + \sqrt{5}$$

(202)  $8 - 2\sqrt{15}$  નું વર્ગમૂળ શોધો.

- (A)  $\sqrt{5} + \sqrt{3}$  (B)  $\sqrt{5}$   
(C)  $\sqrt{5} - \sqrt{3}$  (D)  $2 + \sqrt{5}$

**Solution :**

- ⇒ 8 ના એવા ભાગ પાડો કે જેનો ગુણાકાર 15 થવો જોઈએ.  
⇒ રકમમાં (-) ની નિશાની હોય તો નાની સંખ્યા આગળ લખવી.  
(- હોય તો - લખવા.)

$$\sqrt{5} - \sqrt{3}$$

$$\Rightarrow \sqrt{(\sqrt{5})^2 + (\sqrt{3})^2 - 2\sqrt{5}\sqrt{3}}$$

$$\Rightarrow (a - b)^2 = a^2 + b^2 - 2ab$$

$$\Rightarrow (\sqrt{5} - \sqrt{3})^{2 \times \frac{1}{2}} = \sqrt{5} - \sqrt{3}$$

$$\therefore \text{(C) } \sqrt{5} - \sqrt{3}$$

(203)  $\frac{\sqrt{7}-\sqrt{5}}{\sqrt{7}+\sqrt{5}} = ?$

- (A)  $6 - \sqrt{35}$  (B)  $6 + \sqrt{35}$   
(C)  $3 - \sqrt{32}$  (D)  $3 + \sqrt{32}$

**Solution :**

$$\Rightarrow \frac{\sqrt{7}-\sqrt{5}}{\sqrt{7}+\sqrt{5}} \times \frac{\sqrt{7}-\sqrt{5}}{\sqrt{7}-\sqrt{5}} = \frac{(\sqrt{7}-\sqrt{5})^2}{(\sqrt{7})^2 - (\sqrt{5})^2}$$

$$= \frac{(\sqrt{7})^2 - 2\sqrt{7}\sqrt{5} + (\sqrt{5})^2}{7 - 5}$$

$$= \frac{7 - 2\sqrt{35} + 5}{2}$$

$$= \frac{12 - 2\sqrt{35}}{2}$$

$$= \frac{2(6 - \sqrt{35})}{2}$$

$$= 6 - \sqrt{35}$$

$$\therefore \text{(A) } 6 - \sqrt{35}$$

(204)  $\frac{\sqrt{7}+\sqrt{5}}{\sqrt{7}-\sqrt{5}} = ?$

- (A)  $6 - \sqrt{35}$  (B)  $6 + \sqrt{35}$   
(C)  $3 - \sqrt{32}$  (D)  $3 + \sqrt{32}$

**Solution :**

$$\Rightarrow \frac{(\sqrt{7}+\sqrt{5})^2}{(\sqrt{7})^2 - (\sqrt{5})^2} = \frac{12 + 2\sqrt{35}}{2}$$

$$= \frac{2(6 + \sqrt{35})}{2}$$

$$= 6 + \sqrt{35}$$

$$\therefore \text{(B) } 6 + \sqrt{35}$$

(205)  $\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}} = ?$

- (A)  $2 + \sqrt{15}$  (B)  $2 - \sqrt{15}$   
(C)  $4 + \sqrt{15}$  (D)  $4 - \sqrt{15}$

**Solution :**

$$\Rightarrow \frac{(\sqrt{5}+\sqrt{3})^2}{(\sqrt{5})^2 - (\sqrt{3})^2} = \frac{8 + 2\sqrt{15}}{2}$$

$$= \frac{2(4 + \sqrt{15})}{2}$$

$$= 4 + \sqrt{15}$$

$$\therefore \text{(C) } 4 + \sqrt{15}$$

(206)  $\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}} + \frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}} = ?$

- (A) 5 (B) 6 (C) 7 (D) 8

**Solution :**

$$= \frac{8}{2} + \sqrt{15} + \frac{8}{2} - \sqrt{15}$$

$$= \frac{8}{2} + \frac{8}{2}$$

$$= \frac{8+8}{2}$$

$$= \frac{16}{2}$$

$$= 8$$

$$\therefore \text{(D) } 8$$

(207)  $\frac{\sqrt{7}+\sqrt{5}}{\sqrt{7}-\sqrt{5}} + \frac{\sqrt{7}-\sqrt{5}}{\sqrt{7}+\sqrt{5}} = ?$

- (A) 10 (B) 11 (C) 9 (D) 12

**Solution :**

$$= \frac{6}{2} + \sqrt{35} + \frac{6}{2} - \sqrt{35} = 12$$

$$\therefore \text{(D) } 12$$





$$\begin{array}{r} 49 \swarrow 1 \\ 64 \swarrow 14 \end{array} 50 \quad \frac{50}{7} \Rightarrow \begin{array}{r} 7.14 \\ 7.00 \\ \hline 14.14 \end{array}$$

$$\therefore \frac{14.14}{2} = 7.07$$

$$\therefore \frac{\sqrt{50}}{10} = \frac{7.07}{10} = 0.707$$

$$\therefore (A) 0.707$$

(215) 0.8 નું વર્ગમૂળ શોધો.

- (A) 0.964 (B) 0.694  
(C) 0.894 (D) 0.984

Solution :

$$\therefore \frac{8 \times 10}{10 \times 10} = \sqrt{\frac{80}{100}} = \frac{\sqrt{80}}{10}$$

$$\begin{array}{r} 64 \swarrow 16 \\ 81 \swarrow 1 \end{array} 80 \quad \frac{80}{9} \Rightarrow \begin{array}{r} 8.88 \\ 9.00 \\ \hline 17.88 \end{array}$$

$$\therefore \frac{17.88}{2} = 8.94$$

$$\therefore \frac{\sqrt{80}}{10} = \frac{8.94}{10} = 0.894$$

$$\therefore (C) 0.894$$

(216) 0.6 નું વર્ગમૂળ શોધો.

- (A) 0.944 (B) 0.874  
(C) 0.894 (D) 0.775

Solution :

$$\therefore \frac{6 \times 10}{10 \times 10} = \sqrt{\frac{60}{100}} = \frac{\sqrt{60}}{10}$$

$$\begin{array}{r} 49 \swarrow 11 \\ (64) \swarrow 4 \end{array} 60 \quad \frac{60}{8} \Rightarrow \begin{array}{r} 7.5 \\ 8.0 \\ \hline 15.5 \end{array}$$

$$\therefore \frac{15.5}{2} = 7.75$$

$$\therefore \frac{7.75}{10} = 0.775$$

$$\therefore (D) 0.775$$

## Type # 22

(217) નીચે આપેલ અપૂર્ણાંકમાંથી કયો અપૂર્ણાંક સૌથી નાનો છે.

- (A)  $\frac{51}{57}$  (B)  $\frac{23}{26}$   
(C)  $\frac{31}{35}$  (D)  $\frac{89}{101}$

Solution :

$$\frac{51}{57} \curvearrowright 6 \times 2 \quad \frac{23}{26} \curvearrowright 3 \times 4 \quad \frac{31}{35} \curvearrowright 4 \times 3 \quad \frac{89}{101} \curvearrowright 12$$

6, 3, 4, 12

102 92 93 89

$$\therefore (D) \frac{89}{101}$$

(218) નીચે આપેલ અપૂર્ણાંકને ચડતા ક્રમમાં ગોઠવો.

1.  $\frac{25}{31}$  2.  $\frac{11}{14}$  3.  $\frac{15}{19}$  4.  $\frac{41}{53}$

- (A) 1, 2, 3, 4 (B) 1, 4, 3, 2  
(C) 4, 2, 3, 1 (D) 2, 3, 4, 1

Solution :

$$6 \curvearrowright \frac{25}{31} \quad 3 \curvearrowright \frac{11}{14} \quad 4 \curvearrowright \frac{15}{19} \quad \frac{41}{53} \curvearrowright 12$$

1 2 3 4

6 3 4 12 (12)

50 44 45 41

$$\therefore \frac{41}{53}, \frac{11}{14}, \frac{15}{19}, \frac{25}{31}$$

$$\therefore (C) 4, 2, 3, 1$$

(219) નીચે આપેલ અપૂર્ણાંકને ચડતા ક્રમમાં ગોઠવો.

1.  $\frac{93}{95}$  2.  $\frac{97}{99}$  3.  $\frac{85}{87}$  4.  $\frac{91}{93}$

- (A) 2, 3, 1, 4 (B) 1, 2, 3, 4  
(C) 3, 4, 1, 2 (D) 4, 2, 3, 1

Solution :

$$\frac{93}{95} \curvearrowright 2 \quad \frac{97}{99} \curvearrowright 2 \quad \frac{85}{87} \curvearrowright 2 \quad \frac{91}{93} \curvearrowright 2$$

$$\therefore (C) 3, 4, 1, 2$$

(220) નીચે આપેલ અપૂર્ણાંકને ચડતા ક્રમમાં ગોઠવો.

1.  $\frac{147}{150}$  2.  $\frac{181}{184}$  3.  $\frac{193}{196}$  4.  $\frac{141}{144}$

- (A) 4, 1, 2, 3 (B) 4, 2, 3, 1  
(C) 1, 2, 3, 4 (D) 2, 3, 4, 1